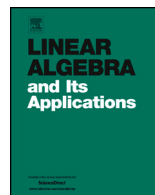




Contents lists available at ScienceDirect

Linear Algebra and its Applications

journal homepage: www.elsevier.com/locate/laa



Gradings and graded identities of null-filiform Leibniz algebras



Lucio Centrone^a, Luís Fertunani^{b,1}, Claudemir Fideles^{b,*,2}

^a *Dipartimento di Matematica, Università degli Studi di Bari Aldo Moro, via Orabona 4, 70125, Bari, Italy*

^b *Department of Mathematics, UNICAMP, 13083-859 Campinas, SP, Brazil*

ARTICLE INFO

Article history:

Received 21 July 2025

Received in revised form 3 November 2025

Accepted 4 November 2025

Available online 10 November 2025

Submitted by M. Brešar

MSC:

17A32
(13A02
16W50
16R10)

Keywords:

Graded identities

Specht property

Null-filiform Leibniz algebras

ABSTRACT

We classify gradings on null-filiform Leibniz algebras up to equivalence over arbitrary fields. Furthermore, we provide a basis for the graded identities and determine a basis of the relatively free algebra. As a consequence, we establish that the ideal of all graded identities of null-filiform Leibniz algebras satisfy the Specht property. Finally, we extend these results to infinite-dimensional analogs of null-filiform Leibniz algebras.

© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

* Corresponding author.

E-mail addresses: lucio.centrone@uniba.it (L. Centrone), l245102@dac.unicamp.br (L. Fertunani), fideles@unicamp.br (C. Fideles).

¹ L. Fertunani was partially supported by FAPESP grant No. 2023/04011-9 and by PhD grant (Finance Code 001) from CAPES, Brazil.

² C. Fideles was partially supported by CNPq grant No. 305651/2021-8 and FAPESP grant No. 2023/04011-9.

1. Introduction

In 1993, J-L. Loday [23] introduced a non-antisymmetric version of Lie algebras, known as *Leibniz algebras*, characterized by the Leibniz identity

$$x(yz) = (xy)z - (xz)y. \quad (1)$$

However, Leibniz algebras were first introduced by A. M. Bloh [8] in the 1960s under the name of *D-algebras*, highlighting their close relationship with derivations. This concept generalizes Lie algebras by relaxing the anti-symmetry condition on the Lie bracket, allowing for a broader class of algebraic structures. Since then, Leibniz algebras have been extensively studied and applied in various branches of mathematics and theoretical physics. For further information, we refer the reader to [6,24] and the references therein.

Among the most celebrated achievements in the theory of algebras with polynomial identities is the theory developed by A. Kemer [2,21], for associative p.i.-algebras over a field of characteristic zero. This theory led him to a positive solution to the long-standing *Specht problem*, i.e., that the ideal of identities of every associative algebra over a field of characteristic zero is finitely generated as a *T*-ideal. Graded algebras and graded polynomial identities play an important role in the proof of Kemer's results; for example, one of the key steps is the deep result that any associative p.i.-algebra satisfies the same identities as the Grassmann envelope of a finite dimensional superalgebra.

Graded algebras have been extensively studied in the last decades. The reader may find quite a large list of references about gradings on algebras in the monographs [4,14]. Gradings on nonassociative algebras by abelian groups have played a crucial role on Lie algebras and superalgebras. This is evident from the root space decomposition of semisimple Lie algebras over the complex field, which constitutes an essential aspect in the Killing-Cartan classification of these algebras.

According to the Killing-Cartan classification, the isomorphism classes of simple Lie algebras, over an algebraically closed field of characteristic zero, are in one-to-one correspondence with irreducible root systems. Such classification is well known and consists of the Classical and Exceptional Lie algebra. It is well known that a Leibniz algebra decomposes into a semidirect sum of the solvable radical and a semisimple Lie algebra [7]. Regarding the nilpotency index, recall that in the theory of Lie algebras, a finite-dimensional filiform Lie algebra over a field K is a nilpotent Lie algebra L whose index of nilpotency is maximal, that is, $\text{nilindex}(L) = \dim(L) - 1$. In the case of Leibniz algebras the class of null-filiform Leibniz algebras and their properties were originally introduced in [3]. The maximal index of nilpotency of an n -dimensional Leibniz algebra is equal to $n + 1$, whereas, the maximal index of nilpotency of an n -dimensional Lie algebra is n . Recently, in [9], it was proved that if there is an abelian grading on a null-filiform Leibniz algebra, then such a group is cyclic.

Although the algebras mentioned above are important in terms of their structure, their identities are far from being understood. The study of polynomial identities for Leibniz al-

gebras was started by Drensky and Piacentini Cattaneo in [13]. Abanina and Mishchenko [1] studied the variety of Leibniz algebras determined by the identity $x(y(z t)) \equiv 0$ and proved many properties similar to the variety of abelian-by-nilpotent Lie algebras of class 2. Several years later Mishchenko and Valenti [27,28] studied the variety of Leibniz algebras defined by the identity $x(y z)(t w) \equiv 0$ and provided a description of multilinear identities in the language of Young diagrams through representation theory of symmetric groups. On polynomial identities and gradings on null-filiform Leibniz algebras, we cite here two important papers. In the first, Centrone and Zarger [10] gave a complete description of the variety of null-filiform Leibniz algebras under the action of Hopf algebras and under the action of a Taft algebra too. In the second, over an infinite field, de Melo and Souza [26] provided the identities and image polynomials on null-filiform Leibniz algebras. Thus, a complete description of the graded identities of null-filiform Leibniz algebras is not known so far.

Returning to the Specht property, we recall that for a wide range of groups and algebras the analogue of the Specht problem was investigated and solved. Iltyakov [19] proved Kemer's theorem to large classes of Lie algebras which include the finite dimensional ones. We also mention the paper by Oates and Powell [30], which proved that the variety of groups generated by a finite group admits a finite basis of its identities. It is also known that the variety generated by a finite ring satisfies the Specht property; this was obtained independently by Kruse [22] and by Lvov [25]. Bahturin and Ol'sanskiĭ proved in [5] that the identities of a finite Lie ring or Lie algebra also admit a finite basis of their identities. Note that every variety of nilpotent algebras has the Specht property. We also cite three more articles on this topic. Naurazbekova and Umirbaev [29] proved that in characteristic 0 any proper subvariety of the variety of dual Leibniz algebras (also called Zinbiel algebras) is nilpotent. Consequently, they have the Specht property. In the second one, Centrone and Zargheh [10] proved, under additional conditions, that a variety generated by a null-filiform Leibniz algebra graded by a group G satisfies the Specht property if and only if G is finite. These positive results hold in characteristic 0. And at the last one, Correa and Koshlukov [11] showed that every ideal of graded identities in the free graded Lie algebra that contains the graded identities of the Lie algebra of the $n \times n$ upper triangular matrices, denoted by $UT_n^{(-)}(K)$, is finitely generated.

We recall that the Specht property does not hold in general. Vaughan-Lee [33] and Drensky [12] proved that Lie algebras over fields of positive characteristic do not satisfy the Specht property. The examples in the case of associative algebras turned out to be much more intricate; such examples were provided simultaneously and independently by Grishin, Belov and Shchigolev [17,20,32]. Additionally, over an infinite field of characteristic 2, Correa and Koshlukov [11] constructed explicitly an ideal of graded identities containing those of $UT_3^{(-)}(K)$ which is not finitely generated as an ideal of graded identities.

It turns out it is an interesting problem to investigate more closely the structure of the graded polynomial identities of null-filiform Leibniz algebras. In this paper, we will give a complete description of the graded identities of null-filiform Leibniz algebras graded by

an abelian group and provide a basis for the relatively free algebra as well. Furthermore, we also prove that each of these T -ideals satisfies the Specht property. We highlight the fact we are working over an arbitrary field.

2. Preliminaries

In this paper K denotes a field. For any arbitrary algebra L , we define its lower central series

$$L^{(1)} = L, \quad L^{(k+1)} = L^{(k)}L, \quad k \geq 1.$$

We say that L is *nilpotent* if $L^{(m)} = 0$ for some $m > 0$ and we call *nilpotency index* of L the minimal m so that $L^{(m)} = 0$. Indeed, the nilpotency index of any n -dimensional nilpotent algebra is at most $n + 1$.

Remark 2.1. Following Ayupov and Omirov [3], we will use this notion throughout the paper. Thus, a nilpotent algebra for us is actually a **right nilpotent** one, for the sake of simplicity in notation. It is well known that, in general, the notions of left, right, and strong nilpotency in Leibniz algebras do not coincide. For instance, consider the two-dimensional Leibniz algebra generated by x and y with multiplication rules

$$[x, x] = [y, y] = [y, x] = 0, \quad [x, y] = x.$$

It is easy to verify that this algebra is left nilpotent but not right nilpotent.

Definition 2.2. [3, Definition 3] An n -dimensional Leibniz algebra L is a null-filiform algebra if $\dim L^{(i)} = (n + 1) - i$, $1 \leq i \leq n + 1$.

The following plays a crucial role in our work: it asserts that, up to isomorphism, there exists only one null-filiform Leibniz algebra in each dimension.

Lemma 2.3. [3, Lemma 1] *In every null-filiform Leibniz algebra of dimension n , there is a basis $\{e_1, e_2, \dots, e_n\}$ with the following multiplication table:*

$$e_i e_1 = e_{i+1}, \quad e_j e_k = 0, \tag{2}$$

for $1 \leq i \leq n - 1$, $j \geq 1$ and $k \geq 2$.

The algebra arising from the last result will be denoted by $\mathcal{L}_0^n$. Of course, a null-filiform Leibniz algebra of dimension $n > 1$ cannot be a Lie algebra.

Let L be an algebra (not necessarily a Leibniz algebra) and let G be a group. A G -grading on L is a vector space decomposition

$$\Gamma: L = \oplus_{g \in G} L^g \quad (3)$$

such that $L^g L^h \subseteq L^{gh}$, for all $g, h \in G$. In this case one simply says that L is G -graded. The subspaces L^g represent the *homogeneous components* of the grading, and a non-zero element a of L is *homogeneous* if $a \in L^g$ for some $g \in G$; it is denoted by $\|a\|_G = g$ (or simply $\|a\| = g$ when the group G is clear from the context). The support of the grading is the set $\text{supp } \Gamma = \{g \in G \mid L^g \neq 0\}$. In particular, a grading Γ on L is said to be of *full support* if $\text{supp } \Gamma = G$. A subalgebra (an ideal, a subspace) B of L is a *graded subalgebra* (respectively *ideal*, *subspace*) if $B = \oplus_{g \in G} (L^g \cap B)$. If L, L' are G -graded algebras, an algebra homomorphism $\varphi: L \rightarrow L'$ is a homomorphism of graded algebras if $\varphi(L^g) \subseteq (L')^g$ for all $g \in G$. Such homomorphisms are called G -graded ones.

There are numerous fascinating and significant examples of graded algebras, including associative, Lie, and others. In this paper, we focus specifically on certain gradings of null-filiform Leibniz algebras of dimension $n > 1$.

Notice that Lemma 2.3 gives us that any null-filiform Leibniz algebra $\mathcal{L}_0^n$ of dimension n can be identified with a Leibniz algebra generated by a single element, denoted by e_1 . Moreover, $\{e_i \mid i = 1, \dots, n\}$ forms a basis for $\mathcal{L}_0^n$, and the multiplication table is defined according to (2). The algebra $\mathcal{L}_0^n$ admits a natural $\mathbb{Z}$ -grading, where e_r is homogeneous of degree r for $r \in \{1, \dots, n\}$, and all other homogeneous components are set to zero. Furthermore, we can define a $\mathbb{Z}$ -grading on $\mathcal{L}_0^n$ by specifying that e_r lies in $(\mathcal{L}_0^n)^s$ if and only if $r = qi + s$, for some $q \in \mathbb{Z}$. Over a field of characteristic zero, all abelian group gradings of $\mathcal{L}_0^n$ were classified in [9]. The study of $\mathcal{L}_0^n$ is particularly relevant for understanding how gradings affect the structure of algebras and the identities they satisfy.

Let $\{X^g\}_{g \in G}$ be a family of disjoint sets $X^g = \{x_1^g, x_2^g, \dots\}$, we denote by $\mathcal{L}\langle X^G \rangle$ the free G -graded Leibniz algebra, freely generated by the set $X^G = \cup_{g \in G} X^g$. This algebra has a natural G -grading assuming the elements of X^g homogeneous of degree g . Consequently, the homogeneous component $(\mathcal{L}\langle X^G \rangle)^g$ is the subspace spanned by all monomials $x_{i_1}^{g_1} \cdots x_{i_m}^{g_m}$ that stands for a monomial with all possible brackets arrangements such that $g_1 \cdots g_m = g$. The elements of $\mathcal{L}\langle X^G \rangle$ are called *G -graded polynomials* (or simply *polynomials*). Actually, by Leibniz identity (1), any polynomial in $\mathcal{L}\langle X^G \rangle$ can be expressed as a linear combination of left-normed monomials, i.e., monomials of the form $(\dots((x_{i_1}^{g_1} x_{i_2}^{g_2}) x_{i_3}^{g_3}) \dots x_{i_{i-1}}^{g_{i-1}})$. For this reason, we will omit the brackets in left-normed monomials. For example, the above monomial will be denoted simply by $x_{i_1}^{g_1} \cdots x_{i_m}^{g_m}$. Furthermore, the expression $f(x_1^{g_1}, \dots, \hat{x}_j^{g_j}, \dots, x_m^{g_m})$ indicates $x_j^{g_j}$ could not appear. The degree of a polynomial f in $x_i^{g_i}$, denoted by $\deg_{x_i^{g_i}} f$, counts how many times the variable $x_i^{g_i}$ appears in the monomials of f , and it is defined in the ordinary way. The definitions of multilinear and multihomogeneous polynomials follow the natural way. Let G be an abelian group, and let $\mathcal{V}$ be the variety of Leibniz algebras equipped with a G -grading. The existence of the free algebra in $\mathcal{V}$ follows from general arguments, see for example [34, Theorem 2]. These arguments yield that two free algebras in $\mathcal{V}$ are isomorphic if and only if their free generating sets have the same cardinality.

Let $\Gamma: L = \bigoplus_{g \in G} L^g$ be a G -graded Leibniz algebra. An *admissible substitution* for the polynomial $f(x_1^{g_1}, \dots, x_n^{g_n})$ in $\mathcal{L}\langle X^G \rangle$ is an n -tuple $(l_1, \dots, l_n) \in L^{g_1} \times \dots \times L^{g_n}$. If $f(l_1, \dots, l_n) = 0$ for every admissible substitution $(l_1, \dots, l_n)$ we say that $f(x_1^{g_1}, \dots, x_n^{g_n})$ is a *graded identity* for L . An ideal invariant under the graded endomorphisms of $\mathcal{L}\langle X^G \rangle$ is called T_G -ideal. In this case, the set of graded identities for any graded Leibniz algebra L equipped with a grading Γ , denoted by $T_G(\Gamma)$, is a T_G -ideal. The intersection of a family of T_G -ideals in $\mathcal{L}\langle X^G \rangle$ is a T_G -ideal; A subset $\mathcal{P} \subseteq \mathcal{L}\langle X^G \rangle$ is a *basis for the T_G -ideal $T_G(\Gamma)$* if $T_G(\Gamma)$ is the intersection of all T_G -ideals in $\mathcal{L}\langle X^G \rangle$ which contain $\mathcal{P}$; this T_G -ideal is denoted by $\langle \mathcal{P} \rangle_{T_G}$. Moreover, it is well known that every T_G -ideal is generated by its *regular polynomials*. Recall that a polynomial $f = f(x_1, \dots, x_n) \in \mathcal{L}\langle X^G \rangle$ is regular if x_i belongs to every summand of f for every $i = 1, \dots, n$. Furthermore, if the ground field has characteristic 0, every T_G -ideal is generated by its multilinear polynomials whereas over an infinite field of positive characteristic one has to take into account the multihomogeneous polynomials instead of the multilinear ones.

3. Gradings and graded identities for $\mathcal{L}_0^n$

In this section, we focus on determining bases for graded identities of null-filiform Leibniz algebras with gradings by arbitrary groups.

Definition 3.1. Let $\Gamma_G: L = \bigoplus_{g \in G} L^g$ and $\Gamma_H: L' = \bigoplus_{h \in H} (L')^h$ be algebras equipped with a grading by the groups G and H , respectively. The graded algebras L and L' are said to be **equivalent** if there exist:

- (1) a bijection $\rho: \text{supp } \Gamma_G \rightarrow \text{supp } \Gamma_H$;
- (2) an isomorphism of algebras $\varphi: L \rightarrow L'$ such that, for every $g \in \text{supp } \Gamma_G$, we have

$$\varphi(L^g) = (L')^{\rho(g)}.$$

This definition ensures that the two algebras have identical algebraic structures under their respective gradings, up to the relabeling of the grading groups and a corresponding algebra isomorphism.

Let G be a finite abelian group. It is well known that there exists a duality between G -gradings on a finite dimensional algebra A and G -actions on A . In this case, the ground field is of characteristic 0 and contains a primitive $|G|$ -th root of unity (see [14, Section 1.4] or [16]). Using this framework, Calderón et al. [9] classified all abelian group gradings on $\mathcal{L}_0^n$.

A grading of an algebra is said to be *toral* if it is generated by an automorphism within a torus of the automorphism group of the algebra. Thus, a cyclic grading is a toral grading generated by a single toral element. In this case, the grading is always equivalent to a grading by a cyclic group. Calderón et al. proved the following theorem regarding gradings of $\mathcal{L}_0^n$.

Lemma 3.2. [9, Lemma 11, Theorem 13] Let $\mathcal{L}_0^n$ be a null-filiform Leibniz algebra of dimension n . Then, up to equivalence, the abelian group gradings on $\mathcal{L}_0^n$ are the following:

- (1) the trivial grading given by $\mathcal{L}_0^n = \langle e_1, \dots, e_n \rangle$;
- (2) the $\mathbb{Z}$ -grading given by $\Gamma_{\mathbb{Z}}^n: \mathcal{L}_0^n = \langle e_1 \rangle_1 \oplus \langle e_2 \rangle_2 \oplus \dots \oplus \langle e_n \rangle_n$;
- (3) for any $2 \leq i \leq n-1$, the $\mathbb{Z}_i$ -grading given by $\Gamma_i^n: \mathcal{L}_0^n = L^{\bar{0}} \oplus L^{\bar{1}} \oplus \dots \oplus L^{\overline{i-1}}$, where each homogeneous subspace $L^{\bar{s}}$ is generated by elements e_j , such that $j \equiv s \pmod i$.

In particular, all abelian group gradings on a null-filiform Leibniz algebra are cyclic toral.

Because we deal with a null-filiform Leibniz algebra graded by any group over an arbitrary field, the situation demands a more careful treatment. Although the classification is simpler in this context, it does not follow from a direct application of the duality discussed above. Rather, we develop the required framework and results.

Theorem 3.3. Let $\mathcal{L}_0^n$ be a null-filiform Leibniz algebra of dimension n graded by a group G . Then there exists a basis $\{e_1, e_2, \dots, e_n\}$ with multiplication table given by (2) such that the element e_1 is homogeneous with respect to this grading. In particular, any grading on $\mathcal{L}_0^n$, up to equivalence, is one of those in Lemma 3.2.

Proof. Let G be a group with identity element ϵ and let $\Gamma: L = \bigoplus_{g \in G} L^g$ be a G -grading on $\mathcal{L}_0^n$. First, there exists an element $a \in L^{(1)} \setminus L^{(2)}$ that is homogeneous with respect to this grading. Otherwise, we may consider

$$a = u_1 + \dots + u_s,$$

where each u_i is homogeneous of G -degree g_i , and all of them are zero in the quotient space $L^{(1)}/L^{(2)}$. This means $u_i \in L^{(2)}$ for all i . Consequently, $a \in L^{(2)}$, which contradicts the assumption that $a \in L^{(1)} \setminus L^{(2)}$. Therefore, a is a homogeneous element in the G -grading Γ , and we denote its G -degree by $\|a\|_G = g$. If $b = aa \neq 0$, we set $e_1 := a$ and $e_2 := b$. Notice that such a, b always exist because, otherwise, $L^{(2)} = 0$ that is an absurd if $n > 1$. Inductively, we may define:

$$e_i := \underbrace{e_1 e_1 \cdots e_1}_{i \text{ times}}.$$

Moreover, $e_i e_j = 0$ for $j \geq 2$; for, if $j = 2$, we have

$$e_i e_2 = e_i (e_1 e_1) = (e_i e_1) e_1 - (e_i e_1) e_1 = 0,$$

where the last equality holds by (1); assume the assertion true for $j > 2$, then by induction we have

$$e_i e_{j+1} = e_i (e_j e_1) = (e_i e_j) e_1 - (e_i e_1) e_j = 0.$$

Thus, for each $i = 1, \dots, n$, the element e_i is homogeneous of G -degree g^i .

Now we have the following cases:

- (1) **Fine grading case:** If $g^i \neq \epsilon$ for all $i = 1, 2, \dots, n$, the grading is fine, meaning each homogeneous component has dimension at most 1. This grading is equivalent to the $\mathbb{Z}$ -grading given by

$$\Gamma_{\mathbb{Z}}^n: \mathcal{L}_0^n = \langle e_1 \rangle_1 \oplus \langle e_2 \rangle_2 \oplus \dots \oplus \langle e_n \rangle_n.$$

- (2) **Non-fine grading case:** If $g^i = \epsilon$ for some $1 \leq i < n$, the specific structure of the grading depends on i . If $i = 1$, we obtain the trivial grading. For $2 \leq i < n$, we write $n = mi + r$ with $0 \leq r \leq i - 1$, $r \in \mathbb{N}$. In this case, we have a $\mathbb{Z}_i$ -grading Γ_i^n with the following homogeneous subspaces:

$$L^r = \text{span}_K\{e_r, e_{i+r}, \dots, e_{im+r}\}.$$

Finally, the gradings described above are not equivalent. This follows immediately from the fact that the cardinality of the support of any given grading differs from the cardinality of the support of any other grading, and we are done. $\square$

From the previous result, we remark that the bases for graded identities of $\mathcal{L}_0^n$ with respect to any group, up to equivalence, have already been determined in Lemma 3.2.

Let Γ_G and Γ_H be equivalent gradings on $\mathcal{L}_0^n$. Then, there exists a homomorphism of group $\rho: G \rightarrow H$ that is a bijection from $\text{supp } \Gamma_G$ to $\text{supp } \Gamma_H$, and an isomorphism of algebras $\varphi: \mathcal{L}_0^n \rightarrow \mathcal{L}_0^n$ such that $\varphi(L^g) = L^{\rho(g)}$ for every $g \in \text{supp } \Gamma_G$. Thus, given a polynomial $f = f(x_1^{g_1}, \dots, x_s^{g_s}) \in \mathcal{L}\langle X^G \rangle$, f is a graded identity for $\mathcal{L}_0^n$ with grading Γ_G if and only if its image $\Phi(f)$ is a graded identity for $\mathcal{L}_0^n$ with grading Γ_H , where $\Phi: \mathcal{L}\langle X^G \rangle \rightarrow \mathcal{L}\langle X^H \rangle$ is the homomorphism defined by

$$\Phi(x_i^g) = x_i^{\rho(g)}, \quad \text{for all } g \in G.$$

All in all, the equivalence of the gradings Γ_G and Γ_H ensures that the graded identities of $\mathcal{L}_0^n$ with the grading Γ_G are in one-to-one correspondence with those of Γ_H via the map ρ .

Proposition 3.4. *Let G and H groups and Γ_G and Γ_H be equivalent gradings on $\mathcal{L}_0^n$. If $\mathcal{P}$ is a basis for the T_G -ideal $T_G(\Gamma_G)$ formed by homogeneous elements in the grading, then $\mathcal{P}' \cup N$ forms a basis for the T_H -ideal $T_H(\Gamma_H)$, where*

$$\mathcal{P}' = \{\Phi(f) \mid f \in \mathcal{P}\}$$

and

$$N = \{x \in X^H \mid \|x\|_H \notin \text{supp } \Gamma_H\}.$$

Proof. We will show that $T_H(\Gamma_H) = \langle \mathcal{P}' \cup N \rangle_{T_H}$. Indeed, $\langle \mathcal{P}' \cup N \rangle_{T_H} \subseteq T_H(\Gamma_H)$ holds, then we only need to prove the reverse inclusion. Let g be an element in $T_H(\Gamma_H)$ and assume, without loss of generality, that $g \notin \langle N \rangle_{T_H}$. Then, there exists $f \in T_G(\Gamma_G)$ written as

$$f = \sum_i \alpha_i m_i(f_1^{g_1^i}, \dots, f_s^{g_s^i}),$$

such that $\Phi(f) = g$, where $\alpha_i \in K$, $m_i(x_1^{g_1^i}, \dots, x_s^{g_s^i})$ is a monomial, and $f_j^{g_j^i}$ is an admissible substitution in $\mathcal{L}\langle X^G \rangle$, for each j , with at least one of the $f_j^{g_j^i}$'s belonging to $\mathcal{P}$ (say $f_s^{g_s^i}$), since $\mathcal{P}$ is formed by homogeneous elements in Γ_G . This implies that $\Phi(f_s^{g_s^i}) \in \mathcal{P}'$, and consequently, $g \in \langle \mathcal{P}' \cup N \rangle_{T_H}$ as desired. $\square$

4. Main results

The main goal of this section is to determine bases for the graded identities of null-filiform Leibniz algebras endowed with gradings by cyclic groups. As established in the previous section, this is sufficient to describe any basis of identities for an arbitrary grading of a given Leibniz algebra.

To enhance clarity, we have organized the discussion into subsections, each dedicated to a specific grading.

When the G -degree of a variable in $\mathcal{L}\langle X^G \rangle$ is clear, it will be omitted.

4.1. Fine grading

The purpose of this subsection is to determine bases for the graded identities of null-filiform Leibniz algebras $\mathcal{L}_0^n$ equipped with a fine grading. By Theorem 3.4, it is enough to consider the grading on $\mathcal{L}_0^n$ by the group $\mathbb{Z}$, denoted by $\Gamma_{\mathbb{Z}}^n$.

It is easy to see that the polynomials

$$x^i, \quad i \notin \{1, 2, \dots, n\}, \quad (4)$$

$$x_1^i x_2^j, \quad 1 \leq i \leq n, \quad 2 \leq j \leq n, \quad (5)$$

are $\mathbb{Z}$ -graded identities of $\mathcal{L}_0^n$.

We need the following technical lemmas.

Lemma 4.1. *Let $M = x_{i_0} x_{i_1} \cdots x_{i_l}$ be a monomial in $\mathcal{L}\langle X_{\mathbb{Z}} \rangle$. If M does not belong to $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ then, modulo the $T_{\mathbb{Z}}$ -ideal generated by the polynomials (4) and (5), it can be written so that $1 \leq \|x_{i_0}\| \leq n$, $\|x_{i_k}\| = 1$ for each $k = 1, \dots, l$ and $\|x_{i_0}\| + l \leq n$.*

Proof. Notice that if $M \notin T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$, then $1 \leq \|x_{i_0}\| \leq n$ and $\|x_{i_1}\| = 1$. Now, suppose there exists k such that $1 < k \leq n$ and $\|x_{i_k}\| \neq 1$. Without loss of generality, we

can assume $k = l$ and $\|x_{i_1}\| = \cdots = \|x_{i_{l-1}}\| = 1$. Then $M' = x_{i_0}x_{i_1}\cdots x_{i_{l-1}}$ is a monomial in $\mathcal{L}\langle X_{\mathbb{Z}} \rangle$ that is not a graded identity for $\mathcal{L}_0^n$. This implies $1 < \sum_{r=0}^{l-1} \|x_{i_r}\| = \|x_{i_0}\| + (l-1) \leq n$. Thus, $M \in T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ since M follows from (5) which is absurd and the proof follows. $\square$

Lemma 4.2. *For every $a \in \mathbb{Z}$, the polynomial $x^a x_1^1 x_2^1 - x^a x_2^1 x_1^1$ lies in the $T_{\mathbb{Z}}$ -ideal generated by the polynomials (4) and (5).*

Proof. The case $a \notin \{1, 2, \dots, n-2\}$ is straightforward. Therefore, we focus on $1 \leq a \leq n-2$. Using identity (1), we have

$$x^a x_1^1 x_2^1 = x^a (x_1^1 x_2^1) + x^a x_2^1 x_1^1,$$

and the assertion follows since $x^a (x_1^1 x_2^1)$ is a consequence of (5). $\square$

Before proceeding with our next result, we will include in our list the polynomial

$$x_1^1 x_2^1 - x_2^1 x_1^1 \tag{6}$$

It is evident that the aforementioned polynomial lies in $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$.

Let J denote the $T_{\mathbb{Z}}$ -ideal generated by the polynomials in (4), (5) and (6).

Recall that a set S of (graded) polynomials is called an independent set of (graded) identities if neither one of its element lies in the ideal of (graded) identities generated by the remaining ones. Thus, for all algebra L , a basis S of (graded) identities of L is said to be *minimal* if it is an independent set.

Proposition 4.3. *Over a ground field K with $|K| > n$, the graded identities*

$$x^i, \quad x_1^j x_2^k \quad \text{and} \quad x_1^1 x_2^1 - x_2^1 x_1^1,$$

where $i \notin \{1, 2, \dots, n\}$, $1 \leq j \leq n$, $2 \leq k \leq n$, and $k+j \leq n$, form a minimal basis of graded identities for the null-filiform Leibniz algebra of dimension n , equipped with a grading by the group $\mathbb{Z}$.

Proof. Indeed, $x_1^i x_2^j$, with $i+j > n$, is a consequence of the graded identity x^{i+j} . Now, let us verify that the remaining elements constitute a minimal basis of graded identities for $\mathbb{Z}$ -graded algebra $\mathcal{L}_0^n$.

Due to the fact that $\mathcal{L}_0^n$ is a nilpotent algebra of index n , we can focus only on the multihomogeneous elements in $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$. This is because if $\deg f \leq n$, then f is a consequence of its multihomogeneous component. See for instance [15, Theorem 1.3.2] and the comment following it.

Assume that $f \in T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ is a multihomogeneous polynomial. By Lemmas 4.1 and 4.2, f is equivalent modulo J , up to a scalar factor, to the monomial

$$x_{i_1} \cdots x_{i_l} = x_0 x_1 \cdots x_1 x_2 \cdots x_{k-1} \underbrace{x_k \cdots x_k}_{m_i^k \text{ times}} x_{k+1} \cdots x_s,$$

where $x_1, \dots, x_s$ have $\mathbb{Z}$ -degree 1. We claim that $f \in T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ if and only if $f \in J$ and the result follows. We prove the claim by induction on the length l of f . The result is obvious for $l = 2$, so we suppose $l \geq 3$. If $f' = x_{i_1} \cdots x_{i_{l-1}} \in T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$, then it lies in J by the induction hypothesis, and hence $f \in J$. We assume now that $f' \notin T_{\mathbb{Z}}(\mathcal{L}_0^n)$. Otherwise, we take $a = \|f'\|$, then the result of every admissible substitution in f' is a scalar multiple of e_a . Therefore, $f \in T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ if and only if $x^a x_l \in T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$. By the case $l = 2$, we have that the monomial $x_1^a x_2^{a_l}$ lies in $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ if and only if $x_1^a x_2^{a_l} \in J$, hence f lies in J .

Finally, the proof will be completed by constructing a $\mathbb{Z}$ -graded Leibniz algebra L that satisfies all identities (4), (5), and (6) but one. To begin, let us choose an integer d such that $d \notin \{1, 2, \dots, n\}$. Define $L = L_1^d$ as the 1-dimensional Lie algebra over K . The algebra $L = \bigoplus_{i \in \mathbb{Z}} (L_1^d)^i$ is $\mathbb{Z}$ -graded with i -th homogeneous component H_i equal to L_1^d if $i = d$ and 0 otherwise. It is clear that L satisfies the graded identities (5) and (6) as well as all graded identities x^c where $c \neq d$ but it does not satisfy the identity x^d . Next, let a and b be two integers such that $1 \leq a \leq n$, $2 \leq b \leq n$ and $a + b \leq n$. We define the Lie algebra of strictly upper triangular 3×3 matrices over K , denoted by $L_{(a,b)} = UT(3, K)$. Now, we introduce a $\mathbb{Z}$ -grading on $L_{(a,b)}$ as follows:

- (1) if $a \neq b$, we define $L_{(a,b)}^k = 0$ for all $k \neq a, b$ and $a + b$; $L_{(a,b)}^a$ is spanned by E_{12} , $L_{(a,b)}^b$ is spanned by E_{23} , and $L_{(a,b)}^{a+b}$ is spanned by E_{13} ;
- (2) if $a = b$, then E_{12} and E_{23} spans $L_{(a,b)}^a$, $L_{(a,b)}^{2a}$ is spanned by E_{13} and $L_{(a,b)}^k = 0$ for all $k \neq a, 2a$.

Here E_{ij} denotes the elementary matrix with entry 1 in the position (i, j) and 0 elsewhere. It is clear that $L_{(a,b)}$ equipped with the previous $\mathbb{Z}$ -grading does not satisfy the graded identity $x_1^a x_2^b$, but it satisfies all other graded identities (4), (5) and (6). Finally, let us consider $UT(3, K)$, equipped with the $\mathbb{Z}$ -grading where L^1 is spanned by E_{12} and E_{23} , L^2 is spanned by E_{13} and $L^k = 0$ for all $k \neq 1$ and 2. Then the unique nonzero product in L is $[E_{12}, E_{23}] = -[E_{23}, E_{12}]$. Consequently, if $n > 1$, all graded identities (4) and (5) are satisfied by L , but it does not satisfy (6) and this concludes the proof. $\square$

In order to simplify the notation we define

$$ab^{(n)} := a \underbrace{bb \cdots b}_{\times n},$$

for all a, b in a Leibniz algebra.

To carry on with the proof over a finite field K of characteristic p and cardinality $|K| = q$, we will include in our list the polynomials

$$(x_0^1)^{(q)}x_1^1 - x_0^1(x_1^1)^{(q)}, \quad (7)$$

$$x_0^a(x_0^1)^{(q)}x_1^1 - x_0^ax_0^1(x_1^1)^{(q)}, \quad (8)$$

where $a > 1$. It is evident that the aforementioned polynomials lie in $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$.

Proposition 4.4. *Over a ground field K with $|K| = q \leq n$, the graded identities*

$$x^i, \quad x_1^j x_2^k, \quad x_1^1 x_2^1 - x_2^1 x_1^1, \quad (x_0^1)^{(q)}x_1^1 - x_0^1(x_1^1)^{(q)} \quad \text{and} \quad x_0^k(x_0^1)^{(q)}x_1^1 - x_0^k x_0^1(x_1^1)^{(q)},$$

where $i \notin \{1, 2, \dots, n\}$, $1 \leq j \leq n$, $2 \leq k \leq n$, and $k + j \leq n$, form a basis for the $\mathbb{Z}$ -graded identities of a null-filiform Leibniz algebra of dimension n .

Proof. Let $f = f(x_0, x_1, \dots, x_n)$ be a regular polynomial in $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$. First, we assume $\|x_0\| \neq 1$. By Lemma 4.1, at most one of the variables in f can have homogeneous degree between 2 and n , and, in this case it must be in the first position of each summand of f . By Lemma 4.2 we can consider all the remaining variables of $\mathbb{Z}$ -degree 1 and reordered according to their lower index. We shall prove the assertion by induction on the degree of the polynomial f . Suppose that f is not multihomogeneous, so, as f is regular, we write $f = \sum_i \alpha_i M_i$ where M_i is written, modulo the $T_{\mathbb{Z}}$ -ideal $\tilde{J}$ generated by J , and identities (7) and (8), as

$$x_0 x_1^{(\alpha_{1,i})} x_2^{(\alpha_{2,i})} \dots x_s^{(\alpha_{s,i})} \quad (9)$$

satisfying $0 \leq \alpha_{k,i} < q$, for all k in $\{2, \dots, s\}$. Furthermore, due to the regularity of f and the characterization of $\Gamma_{\mathbb{Z}}^n$, we suppose, without loss of generality, $\deg M_i = \deg f$.

Of course, the polynomial $y_1 y_2 - y_2 y_1$ and $y^q - y$ form a basis for the polynomial identities of the algebra K . Moreover, the set formed by all monomials $y_{i_1}^{\alpha_1} y_{i_2}^{\alpha_2} \dots y_{i_t}^{\alpha_t}$ satisfying the conditions $i_1 < \dots < i_t$ and $0 \leq \alpha_i < q$ for all i in $\{1, 2, \dots, t\}$, forms a basis of the vector space of the (relatively free) algebra $K[Y]/T(K)$ in the variety defined by the above identities. Here $K[Y]$ denote the free associative algebra freely generated by the set Y .

Notice that f lies in $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ if and only if $f(y_0, y_1, \dots, y_s)$ is an ordinary identity for K . It turns out if, for every M_i in f we have $\alpha_{1,i} < q$, then f lies in $\tilde{J}$. Thus, it remains to consider the case in which there exists at least one M_i in f such that $\alpha_{1,i}$ is greater than or equal to q . In this case, there exists λ_i and r_i in $\mathbb{Z}$ such that $\alpha_{1,i} = \lambda_i q + r_i$, where $0 \leq r_i < q$. Again, the fact that all monomials in f have the same degree implies that $f(y_0, y_1, \dots, y_s)$ will consist of such monomials with different degrees, we have $f \in \tilde{J}$ and we are done. $\square$

By a combination of Propositions 4.3 and 4.4 we have the following result.

Theorem 4.5. *Let K be any field. The $T_{\mathbb{Z}}$ -ideal of every graded identities of the null-filiform Leibniz algebra of dimension n , denoted by $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$, is generated, as a $T_{\mathbb{Z}}$ -ideal, by:*

(i) *The polynomials*

$$x^i, \quad x_1^j x_2^k \quad \text{and} \quad x_1^1 x_2^1 - x_2^1 x_1^1,$$

if $|K| > n$. Moreover, these graded identities form a minimal basis for $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$.

(ii) *If $|K| = q \leq n$, then $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ is generated, as a $T_{\mathbb{Z}}$ -ideal, by the previous polynomials together with the identities*

$$(x_0^1)^{(q)} x_1^1 - x_0^1 (x_1^1)^{(q)} \quad \text{and} \quad x_0^k (x_0^1)^{(q)} x_1^1 - x_0^k x_0^1 (x_1^1)^{(q)}.$$

Here $i \notin \{1, 2, \dots, n\}$, $1 \leq j \leq n$, $2 \leq k \leq n$, and $k + j \leq n$.

As a consequence of the previous result, we provide a description of a basis for the vector space consisting of polynomials in the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}} \rangle / T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$.

Corollary 4.6. *Let S_0 be the set formed by all monomials $\hat{x}_{i_0}^a (x_{i_1}^1)^{(\alpha_1)} \dots (x_{i_l}^1)^{(\alpha_l)}$ with $i_1 < i_2 < \dots < i_l$, $1 < a \leq n$ and $a + \sum_{j=1}^l \alpha_j \leq n$. Over a ground field K , we have that:*

- (i) *If $|K| > n$, then the image of S_0 forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}} \rangle / T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$.*
- (ii) *If $|K| = q \leq n$, then the image of S_0 forms a basis for the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}} \rangle / T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$, where $0 \leq \alpha_k < q$, for all k in $\{2, \dots, l\}$.*

Proof. It suffices to prove that each monomial M is a linear combination of elements in S_0 modulo J , if $|K| > n$, or $\bar{J}$ otherwise, but this was already established in the proof of the previous result. Therefore, the claim holds. $\square$

To conclude this subsection, let $\Gamma_n^i : \mathcal{L}_0^n = \bigoplus_{\bar{g} \in \mathbb{Z}_i} L^{\bar{g}}$ be a $\mathbb{Z}_i$ -grading on $\mathcal{L}_0^n$ satisfying $n < i$. Note that the grading considered here is equivalent to a “fine” grading, which means $\dim L^{\bar{g}} \leq 1$ for all $\bar{g} \in \mathbb{Z}_i$. The proof of the following follows directly from Theorems 3.4 and 4.5.

Corollary 4.7. *Let K be any field. The T_i -ideal of every graded identities of the null-filiform Leibniz algebra of dimension n , denoted by $T_i(\Gamma_n^i)$, is generated, as a T_i -ideal, by:*

(i) *The polynomials*

$$x^{\bar{j}}, \quad x_1^{\bar{k}} x_2^{\bar{\ell}} \quad \text{and} \quad x_1^{\bar{1}} x_2^{\bar{1}} - x_2^{\bar{1}} x_1^{\bar{1}},$$

if $|K| > n$. Moreover, these graded identities form a minimal basis for $T_i(\Gamma_n^i)$.

- (ii) If $|K| = q \leq n$, then $T_i(\Gamma_n^i)$ is generated, as a T_i -ideal, by the previous polynomials together with the identities $(x_0^{\bar{1}})^{(q)}x_1^{\bar{1}} - x_0^{\bar{1}}(x_1^{\bar{1}})^{(q)}$ and $x_0^{\bar{l}}(x_0^{\bar{1}})^{(q)}x_1^{\bar{1}} - x_0^{\bar{l}}x_0^{\bar{1}}(x_1^{\bar{1}})^{(q)}$.

Here $j \notin \{\bar{1}, \bar{2}, \dots, \bar{n}\}$, $1 \leq k \leq n$, $2 \leq \ell \leq n$, and $k + \ell \leq n$.

Furthermore, we can establish another result, which follows the same line of reasoning of Corollary 4.6.

Corollary 4.8. Let S_n^i be the subset of $\mathcal{L}\langle X^{\mathbb{Z}^i} \rangle$ consisting of all monomials

$$\hat{x}_{k_0}^{\bar{a}}(x_{k_1}^{\bar{1}})^{(\alpha_1)} \dots (x_{k_l}^{\bar{1}})^{(\alpha_l)}$$

satisfying $k_1 < \dots < k_l$, $1 < a \leq n$ and $a + \sum_{j=1}^l \alpha_j \leq n$. Over a ground field K , we have that:

- (i) If $|K| > n$, then the image of S_n^i forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}^i} \rangle / T_i(\Gamma_n^i)$.
- (ii) If $|K| = q \leq n$, then the image of S_n^i forms a basis for the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}^i} \rangle / T_i(\Gamma_n^i)$, where $0 \leq \alpha_j < q$, for all j in $\{2, \dots, l\}$.

4.2. Non-fine grading by a nontrivial finite cyclic group

In this subsection, we focus on gradings of the null-filiform Leibniz algebra $\mathcal{L}_0^n$ that are equivalent to gradings by non-trivial finite cyclic groups $\mathbb{Z}_i$ for some positive integer i . The case $n \leq i$ has already been addressed in the previous subsection; therefore, we will only consider the case $2 \leq i < n$. Thus, there exist nonnegative integers q and r such that:

$$n = iq + r,$$

where $r \in \{0, 1, \dots, i-1\}$.

Let $\Gamma_i^n: \mathcal{L}_0^n = \bigoplus_{\bar{g} \in \mathbb{Z}_i} L^{\bar{g}}$ be a $\mathbb{Z}_i$ -grading on $\mathcal{L}_0^n$ with $i < n$. It is worth noticing that the homogeneous component $L^{\bar{g}}$ is spanned, as a vector space, by $e_g, e_{i+g}, \dots, e_{(q-1)+g}$ and e_{iq+g} , if $g \leq r$. In the case $g > r$, the last element can be canceled from the generator list.

Concerning the $\mathbb{Z}_i$ -graded identities on $\mathcal{L}_0^n$, we remark the monomials

$$x_1^{\bar{a}}x_2^{\bar{b}} \tag{10}$$

$$x_0^{\bar{1}}x_1^{\bar{1}}x_2^{\bar{1}}x_3^{\bar{1}} \dots x_{n-i}^{\bar{1}} - x_1^{\bar{1}}x_0^{\bar{1}}x_2^{\bar{1}}x_3^{\bar{1}} \dots x_{n-i}^{\bar{1}} \tag{11}$$

are $\mathbb{Z}_i$ -graded identities of $\mathcal{L}_0^n$, where $1 \leq a \leq i$, $2 \leq b \leq i$.

Lemma 4.9. Let $M = x_{i_0}^{\bar{a}_0} x_{i_1}^{\bar{a}_1} \cdots x_{i_l}^{\bar{a}_l}$ be a monomial in $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle$, where $l > 1$. If $M \notin T_i(\Gamma_i^n)$, then, modulo the T_i -ideal generated by the polynomials (10), it can be written so that $1 \leq a_0 \leq i$ and $a_j = 1$ for each $j = 1, \dots, l$.

Proof. Because $M \notin T_i(\Gamma_i^n)$, we have $1 \leq a_0 \leq i$ and $a_1 \equiv 1 \pmod{i}$. Now, suppose there exists j , with $1 < j \leq l$, and $\bar{a}_j \neq \bar{1}$ in $\mathbb{Z}_i$. We can assume $j = l$ and $a_1 = \cdots = a_{l-1} = 1$. Then, the monomial $M' = x_{i_0}^{\bar{a}_0} x_{i_1}^{\bar{1}} \cdots x_{i_{l-1}}^{\bar{1}}$ is not a graded identity for $\mathcal{L}_0^n$. Note that for each admissible substitution in M' , we have the element

$$\tilde{M}' = \sum_{k \equiv t \pmod{i}} \alpha_k e_k,$$

where $t = a_0 + l - 1$. It is clear that $k \geq 1$ for each k in the above expression. Now, $e_1 \notin L_{a_l}$ implies $M \in T_i(\Gamma_i^n)$ which is a contradiction. $\square$

Lemma 4.10. For any $a \in \mathbb{Z}$, the polynomial $x^{\bar{a}} x_1^{\bar{1}} x_2^{\bar{1}} - x^{\bar{a}} x_2^{\bar{1}} x_1^{\bar{1}}$ and $x_0^{\bar{i}-1} x_1^{\bar{1}} x_2^{\bar{1}} \cdots x_{n-i+2}^{\bar{1}}$ belong to the T_i -ideal generated by the polynomials (10) and (11).

Proof. The first identities of the proof is analogous to that of Lemma 4.2. For the second one, by replacing $x_0^{\bar{1}}$ with $x_0^{\bar{i}-1} y_1^{\bar{1}} y_2^{\bar{1}}$ in (11), we have

$$x_0^{\bar{i}-1} y_1^{\bar{1}} y_2^{\bar{1}} x_1^{\bar{1}} x_2^{\bar{1}} \cdots x_{n-i}^{\bar{1}} - x_1^{\bar{1}} (x_0^{\bar{i}-1} y_1^{\bar{1}} y_2^{\bar{1}}) x_2^{\bar{1}} \cdots x_{n-i}^{\bar{1}} \in I.$$

Since $x_1^{\bar{1}} (x_0^{\bar{i}-1} y_1^{\bar{1}} y_2^{\bar{1}}) x_2^{\bar{1}} \cdots x_{n-i}^{\bar{1}}$ is a consequence of (10), then $x_0^{\bar{i}-1} y_1^{\bar{1}} y_2^{\bar{1}} x_1^{\bar{1}} x_2^{\bar{1}} \cdots x_{n-i}^{\bar{1}}$ lies in I . $\square$

Lemma 4.11. Let $x_1, \dots, x_s$ be variables in $X_{\bar{1}}$. If $n > i$, then any $\mathbb{Z}_i$ -graded identity $f = f(x_1, \dots, x_s)$ of $\mathcal{L}_0^n$, with degree at most $n - i$, is a consequence, modulo the T_i -ideal generated by the graded identities (10) and (11), of graded identities such that all of its summands start with the same variable, let us say x_1 .

Proof. First of all, assume that S is the set consisting of all generators f of $T_i(\Gamma_i^n)$ whose variables have $\mathbb{Z}_i$ -degree equal to 1, such that $f = h + g$, where h is the sum of monomials starting with x_1 , and g is the sum of the remaining ones. Now, let $\Sigma \subseteq S$ be the subset formed by all elements in S such that $g \neq 0$, and assume that $\Sigma \neq \emptyset$.

Let $f = f(x_1, \dots, x_s) \in \Sigma$. Without loss of generality, we can assume $\deg f = n - i$. The case $i = n - 1$ is trivial. Therefore, for a fixed integer i with $2 \leq i < n - 1$, note that both e_1 and e_{i+1} are elements of $L_{\bar{1}}$.

Let I denote the T_i -ideal generated by (10) and (11). Suppose f can be expressed as

$$f = \sum_l \alpha_l M_l \pmod{I}, \quad (12)$$

where each M_l represents linearly independent monomials in $\mathcal{L}\langle X_{\mathbb{Z}_i} \rangle$, modulo I , and $\alpha_l \in K$. Assuming f is regular, the previous lemma shows that each monomial in the decomposition of f can be written as:

$$x_{l_0} x_1 \cdots x_1 x_2 \cdots x_{k-1} \underbrace{x_k \cdots x_k}_{m_i^k \text{ times}} x_{k+1} \cdots x_s, \quad (13)$$

where $l_0 \in \{1, \dots, s\}$ and $m_i^k \geq 0$.

The case where f is multihomogeneous holds, since we consider the evaluation $x_{l_0} = e_1 + e_{i+1}$ and $x_a = e_1$, for all $a \in \{1, 2, \dots, l_0 - 1, l_0 + 1, \dots, s\}$.

Now, let us take f regular, but not multihomogeneous. Let $\ell(f)$ represent the cardinality of the set of nonzero coefficients in the decomposition (12). Assume f *minimal* in the sense that for all h in Σ such that $\ell(h) \neq 0$, we have $\ell(f) \leq \ell(h)$.

Now we take the linear combination of summands of f as in (13) having the same degree of f , and let us denote this polynomial by $\mathcal{M}(f)$. Define $\mathcal{R}(f) = f - \mathcal{M}(f)$, and suppose $\mathcal{M}(f) \notin T_i(\Gamma_i^n)$. Due to the last statement, there exists an admissible substitution for the polynomial $\mathcal{M}(f)$, say $(l_1, \dots, l_s)$, of elements in $L_{\bar{1}}$ such that $\mathcal{M}(f)(l_1, \dots, l_s) \neq 0$. From the definition of $\mathcal{M}(f)$, we have

$$\tilde{\mathcal{M}}(f) = \mathcal{M}(f)(l_1, \dots, l_s) = \alpha_0 e_{n-i} + \alpha_1 e_n.$$

Notice that $\alpha_1 = 0$, since $-\tilde{\mathcal{M}}(f) = \tilde{\mathcal{R}}(f)$ and the degree of $\tilde{\mathcal{R}}(f)$ is less than $n - i$.

Now, we introduce a new variable $x_0 \in X_{\bar{1}}$, and consider

$$\mathcal{R}(f)x_0 = fx_0 - \mathcal{M}(f)x_0.$$

Since $\mathcal{M}(f)x_0$ is linear in x_0 , this polynomial is not a graded identity for $\mathcal{L}_0^n$ if, when evaluated on some $s + 1$ -tuple formed by multiples of e_1 , it is nonzero. The last claim implies that $\mathcal{R}(f)x_0$ is also nonzero when evaluated on the same tuple. In other words,

$$\tilde{\mathcal{R}}(f)e_1 = \alpha_0 e_{n-i+1}.$$

However, such an evaluation is only possible if $\alpha_0 = 0$, due to the length of the monomials contained in $\mathcal{M}(f)$. Moreover, any other evaluation reduces to this one, which leads to a contradiction. Hence, the polynomial $\mathcal{M}(f)$ lies in $T_i(\Gamma_i^n)$, and consequently, $\mathcal{R}(f)$ also lies in $T_i(\Gamma_i^n)$. This implies that, in the generating set, we can add $\mathcal{M}(f)$ and $\mathcal{R}(f)$ in it. In the case $f \neq \mathcal{M}(f)$, the minimality of f in Σ , implies that $\mathcal{M}(f)$ and $\mathcal{R}(f)$ are not in Σ . In this way, we have that f is a consequence of graded identities that starting with the same variable. Otherwise, every monomial in f has degree $n - i$. Then we assume

$$f \equiv \sum_{j=1}^s \alpha_j P_j \pmod{I},$$

where P_j denotes the linear combination of monomials starting with x_j . Without loss of generality, if P_1 is not in $T_i(\Gamma_i^n)$, we can take the following nonzero evaluation: $x_t = \beta_t e_1$, for all $t = 2, \dots, s$, and $x_1 = \omega_0 e_1 + \omega_1 e_{i+1}$. Indeed, we can consider ω_0 and ω_1 to be nonzero. As well as in the multihomogeneous case, we get $\alpha_1 = 0$. We can repeat this process for $j = 2, \dots, s-1$ until we get $f = P_s$ and the proof is complete. $\square$

Notice that the identity (11) cannot be a consequence of the T_i -ideal generated by (10), since we consider a class of filiform non-Lie Leibniz algebras, known as one-parametric filiform Leibniz algebras, defined as follows: let $\mathcal{V}_0^n$ be an n -dimensional algebra over K with a basis $\{e_1, e_2, \dots, e_n\}$ such that the multiplication table of the algebra is given by:

$$e_i e_1 = e_{i+1}, \quad 2 \leq i \leq n-1,$$

with all other products being zero. Recall that $n = iq + r$. This algebra admits a natural $\mathbb{Z}_i$ -grading given by:

$$(\mathcal{V}_0^n)^{\bar{1}} = \text{span}_K \{e_1, e_2, e_{i+2}, e_{2i+2}, \dots, e_{(m-1)i+2}, e_{mi+2}\},$$

and

$$(\mathcal{V}_0^n)^{\bar{a}} = \text{span}_K \{e_{ti+(a+1)} \mid t = 0, 1, 2, \dots, q\}, \quad \text{for all } 2 \leq a \leq i.$$

For further details on this grading, see [9, Section 4]. Notice this algebra is different from the null-filiform Leibniz algebra $\mathcal{L}_0^n$. Actually, in $\mathcal{L}_0^n$, we have $e_1 e_1 = e_2$, whereas in this new algebra, the same product is zero. It is straightforward to verify that (11) is not a graded identity for $\mathcal{V}_0^n$, whereas (10) are.

Henceforth, we will denote by J_i the T_i -ideal generated by the polynomials (10) and (11). Our focus will now be on proving that $J_i = T_i(\Gamma_i^n)$, when $n \geq i$.

As was done in the previous section, we will separate into two cases, according to the cardinality of the ground field. The first case will be when we have a sufficiently large number of elements in K , and thus the following result follows.

Proposition 4.12. *Over an arbitrary field K such that $|K| > n$, the polynomials*

$$x_1^{\bar{a}} x_2^{\bar{b}} \tag{14}$$

$$x_0^{\bar{1}} x_1^{\bar{1}} x_2^{\bar{1}} x_3^{\bar{1}} \cdots x_{n-i}^{\bar{1}} - x_1^{\bar{1}} x_0^{\bar{1}} x_2^{\bar{1}} x_3^{\bar{1}} \cdots x_{n-i}^{\bar{1}}, \tag{15}$$

where $1 \leq a \leq i$, $2 \leq b \leq i$, form a minimal basis for the graded identities of the n -dimensional null-filiform Leibniz algebras equipped with the grading by group $\mathbb{Z}_i$, for some fixed $i = 2, 3, \dots, n$.

Proof. Assume, without loss of generality, that $f = f(x_0, x_1, \dots, x_s)$ is a multihomogeneous polynomial in $T_i(\Gamma_i^n)$. Due to Lemma 4.9 and 4.11, and identity (15), all monomial

in f start with the variable x_0 . Furthermore, as it was done in the previous proof, every multihomogeneous polynomial f , modulo J_i , is equivalent, up to a scalar factor, to the monomial

$$x_0 \cdots x_0 x_1 \cdots x_1 x_2 \cdots x_{k-1} \underbrace{x_k \cdots x_k}_{m^k \text{ times}} x_{k+1} \cdots x_s = x_0 x_0^{(m^0-1)} x_1^{(m^1)} \cdots x_s^{(m^s)},$$

where $x_1, \dots, x_s$ are of degree $\bar{1}$. If $\|x_0\| \neq \bar{1}$, then $m^0 = 1$. Therefore, by Proposition 4.3, it suffices to prove that every $\mathbb{Z}_i$ -graded monomial identity of $\mathcal{L}_0^n$ follows from the graded identities in J_i . We shall prove our claim by induction on the length l of f . Let $f' = x_{i_1} \cdots x_{i_{l-1}}$, we have $f = f' x_{i_l}$. If $f' \in T_i(\Gamma_i^n)$, then it belongs to J_i by the induction hypothesis. Otherwise, if $f' \notin T_i(\Gamma_i^n)$, consider $\bar{a} = \|f'\|$. As $\|x_{i_l}\| = 1$, we have $\bar{a} = \bar{n}$. Lemma 4.10 implies that f lies in J_i . Now, suppose $\|x_0\| = \bar{1}$, we assume the evaluation $x_k = e_1$, for all k , and we are done. Therefore, it remains to verify the minimality of the basis.

Firstly, it was mentioned that (15) is not a consequence of the T_i -ideal generated by (14). Furthermore, let a and b be two integers satisfying $1 \leq a \leq n$, $2 \leq b \leq n$ and $a + b \leq n$. We define the Lie algebra of strictly upper triangular 3×3 matrices over K that is denoted by $L_{(a,b)} = UT(3, K)$. Now, we define a $\mathbb{Z}_i$ -grading on $L_{(a,b)}$ as follows:

- (1) if $a \neq b$, we consider $L_{(a,b)}^k = 0$ for all $k \neq a, b$ and $a + b$; $L_{(a,b)}^a$ is spanned by E_{12} , $L_{(a,b)}^b$ is spanned by E_{23} , and $L_{(a,b)}^{a+b}$ is spanned by E_{13} ;
- (2) if $a = b$, then $L_{(a,b)}^a$ is spanned by E_{12} and E_{23} , $L_{(a,b)}^{2a}$ is spanned by E_{13} , and $L_{(a,b)}^k = 0$ for all $k \neq a, 2a$.

Here, E_{ij} denotes the elementary matrix with 1 in the position (i, j) and 0 elsewhere. It is clear that $L_{(a,b)}$ equipped with the previous $\mathbb{Z}_i$ -grading does not satisfy the graded identity $x_1^a x_2^b$, but it satisfies all other graded identities. And the proof is complete. $\square$

Now we start studying a basis of the $\mathbb{Z}_i$ -graded identities of the n -dimensional null-filiform Leibniz algebras when $|K| = q \leq n$. For, we consider the new identities

$$x_0^{\bar{a}} (x_1^{\bar{1}})^{(q)} x_2^{\bar{1}} - x_0^{\bar{a}} x_1^{\bar{1}} (x_2^{\bar{1}})^{(q)}, \quad (16)$$

$$((x_0^{\bar{1}})^{(q)} x_1^{\bar{1}} - x_0^{\bar{1}} (x_1^{\bar{1}})^{(q)}) x_2^{\bar{1}} \cdots x_{n-i+1-q}^{\bar{1}}, \quad (17)$$

when $a \geq 1$ and $q < n - i$. For $q \geq n - i$, we replace the identity (17) by

$$(x_0^{\bar{1}})^{(q)} x_1^{\bar{1}} - x_0^{\bar{1}} (x_1^{\bar{1}})^{(q)}. \quad (18)$$

It is easy to see the aforementioned polynomials belong to $T_i(\Gamma_i^n)$.

Proposition 4.13. *Over a ground field K with $|K| = q \leq n$, a basis for the graded polynomial identities of the null-filiform Leibniz algebra of dimension n , equipped with a grading*

by the group $\mathbb{Z}_i$, is given by the graded polynomials (14)–(17) if $q < n - i$. Otherwise, the graded polynomial identities are generated by the polynomials (14)–(16) and (18).

Proof. The proof is analogous to that of Lemma 4.2 taking into account we are working with a finite cyclic group. We will write the details, for completeness of proof.

Let $f = f(x_0, x_1, \dots, x_s)$ be a regular polynomial in $T_i(\Gamma_i^n)$. Assume that f is written as

$$f = \sum_{\ell} \alpha_{\ell} M_{\ell},$$

where each monomial M_{ℓ} , module the T_i -ideal J_i , is written as

$$M_{\ell} = x_0 x_1^{(\alpha_{1,\ell})} x_2^{(\alpha_{2,\ell})} \dots x_s^{(\alpha_{s,\ell})}. \quad (19)$$

Here, if $\|x_0\| = \bar{1}$, we omitted this variable in f .

First of all, let us again recall that the set formed by all monomials $y_{i_1}^{\beta_1} y_{i_2}^{\beta_2} \dots y_{i_t}^{\beta_t}$ satisfying the conditions that $i_1 < \dots < i_n$ and $0 \leq \beta_r < q$, for all r in $\{1, 2, \dots, t\}$, forms a basis of the vector space of the relatively free algebra $K[Y]/T(K)$ in the variety defined by the identities $y_1 y_2 - y_2 y_1$ and $y^q - y$. Here, $K[Y]$ denotes the free associative algebra freely generated by the set Y . Thus, f lies in $T_i(\Gamma_i^n)$ implies $f(y_0, y_1, \dots, y_s)$ is an ordinary identity for K . The reciprocal holds if f is linear in x_0 , or $\deg f \geq n - i$.

Denote by $\tilde{J}_i$ the T_i -ideal generated by J_i along with the graded identities (14)–(17) if $q < n - i$. Moreover, if $q \geq n - i$, we consider the graded identities (14)–(16) and (18).

Now, let us analyze the exponents in each monomial M_{ℓ} . Notably, the graded identities (16) imply that for all $k \in \{3, \dots, s\}$, the exponent $\alpha_{k,\ell}$ satisfies $0 \leq \alpha_{k,\ell} < q$. Thus, we need to examine the cases where $\alpha_{1,\ell}$ or $\alpha_{2,\ell}$ is greater than or equal to q . Hence, if for every M_{ℓ} in f with $\alpha_{1,\ell}, \alpha_{2,\ell} < q$, we have that f lies in $\tilde{J}_i$. If there exists some $\alpha_{1,\ell'} \geq q$, but for every M_{ℓ} in f we have $\alpha_{2,\ell} < q$, then there exist integers λ_{ℓ} and r_{ℓ} in $\mathbb{Z}$ such that $\alpha_{1,\ell} = \lambda_{\ell} q + r_{\ell}$, where $0 \leq r_{\ell} < q$. Since all monomials in f have equal degrees, it follows that $f \in \tilde{J}_i$. The case where $\alpha_{2,\ell'} \geq q$ for some ℓ' and $\alpha_{1,\ell} < q$ for all ℓ follows similarly, since the T_i -ideal remains invariant on the free graded algebra $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle$.

Finally, the only remaining case to verify is when there exist monomials M_{ℓ} such that $\alpha_{1,\ell} < q$ and $\alpha_{2,\ell} \geq q$, while others satisfy the reverse inequality. The identities (16) allow us to assume that $\|x_0\| = 1$. Furthermore, if $q \geq n - i$, then (18) ensures that the exponents satisfy $0 \leq \alpha_{k,\ell} < q$ for all $k \in \{2, \dots, s\}$. We obtain the same conclusion when $\deg f \geq n + 1 - i$, by applying identity (17). These situations have already been considered. Thus, we assume $q < n - i$ and $\deg f < n + 1 - i$. Notice that the graded identity

$$(x_1^{\bar{1}})^{(q+1)} x_2^{\bar{1}} - (x_1^{\bar{1}})^{(2)} (x_2^{\bar{1}})^{(q)}$$

follows from the identities (16), which allows us to assume that if $\alpha_{2,\ell} \geq q$ for some ℓ , then M_{ℓ} is linear in x_1 . Thus, if $\deg_{x_1} M_{\ell} > 1$, implies that $\deg_{x_2} M_{\ell} < q$. Because

of these comments, we decompose f as $f = g + h$, where h consists of the sum of all monomials in f satisfying $\alpha_{1,\ell} = 1$, while g is formed by the remaining terms. In other words, f can be expressed in the form

$$f = \underbrace{\sum_{\ell} \lambda_{\ell} x_1 x_2^{(\alpha_{2,\ell})} \cdots x_s^{(\alpha_{s,\ell})}}_{=h} + \underbrace{\sum_{\ell'} \lambda_{\ell'} x_1^{(\alpha_{1,\ell'})} x_2^{(\alpha_{2,\ell'})} \cdots x_s^{(\alpha_{s,\ell'})}}_{=g} \pmod{\tilde{J}_i},$$

where $\alpha_{2,\ell'} \leq q < \alpha_{2,\ell}$ and $\lambda_{\ell}, \lambda_{\ell'} \in K$. In this way, we take the variables $y_1, \dots, y_s$ in $K[Y]/T(K)$, and assume the admissible substitution $x_1 = y_1 e_{i+1}$ and $x_i = y_i e_1$, for each $i = 2, \dots, s$, under these conditions we have

$$0 = f(y_1 e_{i+1}, y_2 e_1, \dots, y_s e_1) = \underbrace{\left(\sum_{\ell} \lambda_{\ell} y_1 y_2^{(\alpha_{2,\ell})} y_3^{(\alpha_{3,\ell})} \cdots y_s^{(\alpha_{s,\ell})} \right)}_{=h(y_1, y_2, \dots, y_s)} e_{i+\deg f}.$$

This implies $h(y_1, y_2, \dots, y_s) \in T(K)$. For any pair of monomials M_{ℓ_1} and M_{ℓ_2} in h , the exponents α_{2,ℓ_1} and α_{2,ℓ_2} must be different. Consequently, the corresponding elements in the quotient ring $K[Y]/T(K)$, for which $\alpha_{i,\ell_1} = \alpha_{i,\ell_2}$ for all $i = 3, \dots, s$, have distinct lengths. Due to this comment, we conclude that each coefficient λ_{ℓ} in h must be zero. The same conclusion is obtained for g if we consider the admissible substitution given by $x_i = y_i e_1$, for each $i = 1, 2, \dots, s$. Therefore, $f \in \tilde{J}_i$, and the proof is complete. $\square$

We are ready to furnish a basis of graded identities for null-filiform Leibniz algebras equipped with the grading by the group $\mathbb{Z}_i$, where $i = 2, \dots, n$.

Theorem 4.14. *Let K be any field and let $\mathcal{L}_0^n$ be the null-filiform Leibniz algebra of dimension n endowed with its natural $\mathbb{Z}_i$ -grading, denoted by Γ_i^n , where $n > i$. We have the next.*

- (i) *If $|K| \geq n$, then the ideal of every graded identities of Γ_i^n , as a T_i -ideal, is generated by the polynomials*

$$\begin{aligned} & x_1^{\bar{a}} x_2^{\bar{b}} \\ & x_0^{\bar{1}} x_1^{\bar{1}} x_2^{\bar{1}} x_3^{\bar{1}} \cdots x_{n-i}^{\bar{1}} - x_1^{\bar{1}} x_0^{\bar{1}} x_2^{\bar{1}} x_3^{\bar{1}} \cdots x_{n-i}^{\bar{1}}. \end{aligned}$$

Moreover, these graded identities form a minimal basis for $T_i(\Gamma_i^n)$.

- (ii) *If $|K| = q < n$ and $q \geq n - i$, then the previous polynomials together with the graded identities $(x_0^{\bar{1}})^{(q)} x_1^{\bar{1}} - x_0^{\bar{1}} (x_1^{\bar{1}})^{(q)}$ and $x_0^{\bar{a}} (x_1^{\bar{1}})^{(q)} x_2^{\bar{1}} - x_0^{\bar{a}} x_1^{\bar{1}} (x_2^{\bar{1}})^{(q)}$ form a basis for $T_i(\Gamma_i^n)$, as a T_i -ideal.*
- (iii) *If $|K| = q < n - i$, then the polynomials given by item (i), together with the graded identities $((x_0^{\bar{1}})^{(q)} x_1^{\bar{1}} - x_0^{\bar{1}} (x_1^{\bar{1}})^{(q)}) x_2^{\bar{1}} \cdots x_{(n-i+1-q)}^{\bar{1}}$ and $x_0^{\bar{a}} (x_1^{\bar{1}})^{(q)} x_2^{\bar{1}} - x_0^{\bar{a}} x_1^{\bar{1}} (x_2^{\bar{1}})^{(q)}$ form a basis for $T_i(\Gamma_i^n)$, as a T_i -ideal.*

Here $1 \leq a \leq i$, $2 \leq b \leq i$.

Proof. All cases have already been considered, since, when $|K| = n$, the graded identities (16) and (18) follow from $x_0^{\overline{i-1}} x_1^{\overline{1}} x_2^{\overline{1}} \cdots x_{n-i+2}^{\overline{1}}$. $\square$

To finish this subsection, we will describe a basis for the vector space of polynomials in the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle / T_i(\Gamma_i^n)$, when $n \leq i$. For this purpose, we establish the following result.

Corollary 4.15. *Let S_i^n be the subset of $\mathcal{L}\langle X_{\mathbb{Z}_i} \rangle$ consisting of all monomials*

$$\hat{x}_{k_0}^{\bar{a}} (x_{k_1}^{\overline{1}})^{(\alpha_1)} \cdots (x_{k_l}^{\overline{1}})^{(\alpha_l)} \quad (20)$$

satisfying $k_1 < \cdots < k_l$, $1 \leq a \leq i$ and $a + \sum_{j=1}^l \alpha_j \leq n$. Over any field K , we have:

- (i) If $|K| \geq n$, then the image of S_i^n forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle / T_i(\Gamma_i^n)$.
- (ii) If $|K| = q < n$ and $q \geq n - i$, then the image of S_i^n forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle / T_i(\Gamma_i^n)$, where $0 < \alpha_j < q$, for all j in $\{2, \dots, l\}$.
- (iii) If $|K| = q < n - i$, then the image of S_i^n forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle / T_i(\Gamma_i^n)$, where $0 < \alpha_j < q$, for all j in $\{3, \dots, l\}$, and only one of the following cases happens:

- (a) $\alpha_1 = 1$, $\alpha_2 \geq q$, $\hat{x}_{k_0}^{\bar{a}}$ does not appear in expression (20) and $\sum_{j=1}^l \alpha_j \leq n - i$, or;
- (b) $\alpha_2 < q$.

Remark 4.16. In the expression (20), if $\hat{x}_{k_0}^{\bar{a}}$ appears in the expression, $a = 1$ and the degree of the monomial is greater than $n - i$, then we also require that $k_0 = k_1$.

4.3. Trivial group

As the final part of this section, we aim to explore analogous results to those established previously for the group $G = \{e\}$. In this case, such graded identities are simply referred to as **identities**. In [26], a finite minimal basis for the set of all identities of $\mathcal{L}_0^n$, denoted by $T(\mathcal{L}_0^n)$, was constructed over an infinite field. The purpose of this subsection is to extend that result to an arbitrary field.

Certainly, the polynomials

$$x_1(x_2x_3) \quad (21)$$

$$x_1x_2x_3 \cdots x_n - x_2x_1x_3 \cdots x_n \quad (22)$$

are identities of the algebra $\mathcal{L}_0^n$. For $|K| = n - 1$, the polynomial

$$x_1^{(n-1)}x_2 - x_1x_2^{(n-1)} \quad (23)$$

is also an identity of $\mathcal{L}_0^n$. If instead $|K| = q < n - 1$, then we disregard the identity (23), and it is easy to verify that the following polynomials lie in $T(\mathcal{L}_0^n)$:

$$(x_0^{(q)}x_1 - x_0x_1^{(q)})x_2 \cdots x_{n-q} \quad \text{and} \quad x_0(x_1)^{(q)}x_2 - x_0x_1(x_2)^{(q)}. \quad (24)$$

Let I denote the T -ideal generated by the polynomials in (21), (22) and (23), in the case where $|K| = n - 1$. On the other hand, if $|K| = q < n - 1$, we replace (23) with the identities in (24).

Lemma 4.17. *The polynomials $x_1x_2x_3 - x_1x_3x_2$ and $x_1 \cdots x_nx_{n+1}$ are in I .*

Proof. The proof for the first identity is analogous to that of Lemma 4.2. The second one is proven in [26]. For completeness, we include its proof here. By replacing x_1 with xy in (22), we have $xyx_2x_3 \cdots x_n - x_2(xy)x_3 \cdots x_n \in I$. Since $x_2(xy)x_3 \cdots x_n$ is a consequence of (21), then $xyx_2x_3 \cdots x_n$ lies in I . $\square$

Now we have the main result of this subsection.

Theorem 4.18. *Let $\mathcal{L}_0^n$ be an n -dimensional null-filiform Leibniz algebra over any field. Then:*

(i) *the polynomials $x_1(x_2x_3)$ and*

$$x_1x_2x_3 \cdots x_n - x_2x_1x_3 \cdots x_n$$

generate the ideal of identities of $\mathcal{L}_0^n$, as a T -ideal, if $|K| \geq n$;

(ii) *if $|K| = n - 1$, then $T(\mathcal{L}_0^n)$ is generated by the previous polynomials together with the identity $x_1^{(n-1)}x_2 - x_1x_2^{(n-1)}$, as a T -ideal;*

(iii) *if $|K| = q < n - 1$, then the polynomials given by item (i), together with the identities $(x_0^{(q)}x_1 - x_0x_1^{(q)})x_2 \cdots x_{n-q}$ and $x_0(x_1)^{(q)}x_2 - x_0x_1(x_2)^{(q)}$ form a basis for $T(\mathcal{L}_0^n)$, as a T -ideal.*

Moreover, in the cases (i) and (ii), these identities form a minimal basis for $T(\mathcal{L}_0^n)$.

Proof. The proof is straightforward for the case $|K| > n$, as it follows the same technique outlined in [26]. Therefore, it remains to address the case $|K| = q \leq n$. Following the argument used in Lemma 4.11, it can be shown that any identity $f = f(x_1, \dots, x_s)$ of $\mathcal{L}_0^n$ is a consequence, modulo (21) and (22), of identities starting with the same variable,

say x_1 . Consequently, all monomials in f have the same degree. Thus, all monomial in f , modulo the ideal generated by I , is written as

$$M_i = x_1^{(\alpha_{1,i})} x_2^{(\alpha_{2,i})} \dots x_s^{(\alpha_{s,i})} \quad (25)$$

satisfying $0 \leq \alpha_{k,i} < q$, for all k in $\{2, \dots, s\}$.

Now, it is easy to see that f lies in $T(\mathcal{L}_0^n)$ implies $f(y_0, y_1, \dots, y_s)$ is an ordinary identity for K . Hence, if $\alpha_{1,i} < q$ for every M_i in f , we have that f lies in I . Thus, it remains to consider the case in which there exists at least one M_i in f such that $\alpha_{1,i}$ must be greater than or equal to q . In this way, there exists λ_i and r_i in $\mathbb{Z}$ such that $\alpha_{1,i} = \lambda_i q + r_i$, where $0 \leq r_i < q$. Now $f \in I$ because all monomials in f have equal degree.

Finally, if $|K| = n - 1$, the identity $x_1^{(n-1)} x_2 - x_1 x_2^{(n-1)}$ cannot be a consequence from the others, since it is not an identity for $\mathcal{L}_0^k \otimes_K F$, for any nontrivial extension F of the ground field. And we are done. $\square$

In addition to the above theorem, one can establish one more result.

Corollary 4.19. *Let S be the subset of $\mathcal{L}\langle X \rangle$ consisting of all monomials*

$$\hat{x}_{i_0} x_{i_1}^{(\alpha_1)} \dots x_{i_l}^{(\alpha_l)}$$

satisfying $i_1 < i_2 < \dots < i_l$ and $\sum_{j=1}^l \alpha_j < n$, if x_{i_0} appears in it, or $\sum_{j=1}^l \alpha_j \leq n$ otherwise. Furthermore, if the degree of the above monomial is n , it yields $i_0 = i_1$. Over a ground field K , we have that:

- (i) *If $|K| > n$, then the image of S forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X \rangle / T(\mathcal{L}_0^n)$.*
- (ii) *If $|K| = n - 1$, then the image of S forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X \rangle / T(\mathcal{L}_0^n)$, where $0 < \alpha_k < n - 1$, for all k in $\{2, \dots, l\}$.*
- (iii) *If $|K| = q < n - 1$, then the image of S forms a basis of the vector space of the relatively free algebra $\mathcal{L}\langle X \rangle / T(\mathcal{L}_0^n)$, where $0 < \alpha_j < q$, for all j in $\{3, \dots, l\}$, and only one of the following cases happens:*

- (a) $\alpha_1 = 1$, $\alpha_2 \geq q$, x_{i_0} does not appear in the above expression and $\sum_{j=1}^l \alpha_j < n$,
or;
- (b) $\alpha_2 < q$.

5. The Specht property

In this section, we deduce the Specht property for the ideal of graded identities for an n -dimensional null-filiform Leibniz algebra when it is equipped with a grading by an abelian group.

Definition 5.1. Let A be a G -graded algebra. The T_G -ideal $T_G(A)$ has the Specht property if every T_G -ideal J such that $T_G(A) \subseteq J$, has a finite basis, that is J is finitely generated as a T_G -ideal.

To begin, we recall briefly the necessary facts concerning the finite basis property for sets. For more details see [18].

A relation $a \leq b$ on a set A is a quasi-order if it is reflexive and transitive. If it also is symmetric, the relation is an order; and if in addition for all a, b either $a \leq b$ or $b \leq a$, it is a linear order. If B is a subset of a quasi-ordered set A , the closure of B is defined as

$$\bar{B} = \{a \in A \mid \text{there exists } b \in B \text{ such that } b \leq a\}.$$

A closed subset is one that coincides with its own closure i.e., $B = \bar{B}$. We assert that the quasi-ordered set A has the finite basis property (f.b.p) if every closed subset of A is the closure of a finite set. Next, we present several alternative definitions of this property.

Theorem 5.2. *The following conditions on a quasi-ordered set A are equivalent.*

- (1) *Every closed subset of A is the closure of a finite subset;*
- (2) *If B is any subset of A , there is a finite B_0 such that $B_0 \subseteq B \subseteq \bar{B}_0$;*
- (3) *Every infinite sequence of elements $\{a_i\}_{i \geq 0}$ of A has an infinite ascending subsequence $a_{i_1} \leq a_{i_2} \leq \dots \leq a_{i_k} \leq \dots$*

The cardinal product of two quasi-ordered sets A and B is the set of couples (a, b) with $a \in A$ and $b \in B$, ordered by $(a_1, b_1) \leq (a_2, b_2)$ if and only if $a_1 \leq a_2$ and $b_1 \leq b_2$.

Theorem 5.3. *Two quasi-ordered sets have f.b.p., so does their cardinal product.*

For example, let k be a positive integer and consider $\mathbb{N}$ endowed with its natural order $\leq$. Let $D(\mathbb{N})$ be the set of all finite sequences of elements of $\mathbb{N}$. Then $D(\mathbb{N})$ is quasi-ordered by the relation $x \leq y$ if there exists a map $\varphi: \mathbb{N} \rightarrow \mathbb{N}$ such that $1 \leq \varphi(k) \leq s$, $\varphi(i) < \varphi(j)$ if $i < j$, and $p_j \leq q_{\varphi(j)}$ for every $j = 1, \dots, k$, where $x = (p_1, \dots, p_k) \leq y = (q_1, \dots, q_s)$. Such map is called *an order preserving injection*. As $(\mathbb{N}, \leq)$ has the finite basis property, so has $D(\mathbb{N})$; this follows directly from [18, Theorem 4.3].

5.1. Infinite cyclic group

Let us consider the case where the abelian group is infinite. Naturally, we encounter an infinite number of identities indexed by elements outside the support. However, for the sake of our results, we will disregard these. Due to Corollary 4.6, we denote

$$\hat{x}_0^a \underbrace{x_1 \cdots x_1}_{\alpha_1} x_2 \cdots x_k = \hat{x}_0^a (x_1)^{(\alpha_1)} \cdots (x_k)^{(\alpha_k)},$$

where $\|x_i\| = 1$, for all $i = 1, \dots, k$. Now we consider the sets defined as follows: for $a = 1$, we consider

$$\mathcal{B}_a = \{(x_1)^{(\alpha_1)} \cdots (x_k)^{(\alpha_k)} \mid 0 \leq \sum_{j=1}^l \alpha_j \leq n, \alpha_i \geq 0, \|x_i\| = 1, \text{ for every } i\},$$

and, if $a \in \{2, \dots, n\}$, we then have

$$\mathcal{B}_a = \{x_0^a(x_1)^{(\alpha_1)} \cdots (x_k)^{(\alpha_k)} \mid 0 \leq \sum_{j=1}^l \alpha_j \leq n - a, \alpha_i \geq 0, \|x_i\| = 1, \text{ for every } i\}.$$

Given $f = \hat{x}_0^a(x_1)^{(\alpha_1)} \cdots (x_k)^{(\alpha_k)} \in \mathcal{B}_a$, we define $d(f) = (\alpha_1, \dots, \alpha_k)$. Observe that $\{d(f) \mid f \in \mathcal{B}_a\} \subseteq D(\mathbb{N})$, and for every f and g in $\mathcal{B}_a$, we said that $d(f) = (\alpha_1, \dots, \alpha_k) \leq (\alpha'_1, \dots, \alpha'_m) = d(g)$ if there exists an order preserving injection.

Lemma 5.4. *Let $f, g \in \mathcal{B}_a$. If $d(f) \leq d(g)$, then g is a consequence of f modulo $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$, that is $g \in \langle f \cup T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n) \rangle_{T_{\mathbb{Z}}}$.*

Proof. Suppose $d(f) = (a_1, \dots, a_k)$ and $d(g) = (a'_1, \dots, a'_m)$. By hypothesis, there exists a subsequence $(a'_{i_1}, \dots, a'_{i_k})$ of $d(g)$ such that $a_j \leq a'_{i_j}$ for every $j \in \{1, 2, \dots, k\}$. Let $f(\hat{x}_0^a, x_{l_1}, \dots, x_{l_k})$ be the polynomial obtained by replacing the variable x_j by x_{l_j} for each $1 \leq j \leq k$ in $f(\hat{x}_0^a, x_1, \dots, x_k)$. Then, by Lemma 4.2 (or equality (6)), we have:

$$f(\hat{x}_0^a, x_{l_1}, \dots, x_{l_k})(x_{l_1})^{(a'_{i_1} - a_1)} \cdots (x_{l_k})^{(a'_{i_k} - a_k)} \equiv \hat{x}_0^a(x_1)^{(a'_{i_1})} \cdots (x_k)^{(a'_{i_1})} \pmod{T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)}.$$

If $\{t_1, \dots, t_{m-k}\} = \{1, \dots, m\} \setminus \{l_1, \dots, l_k\}$, by using Lemma 4.2 (or equality (6)) again, we obtain:

$$\hat{x}_0^a(x_1)^{(a'_{i_1})} \cdots (x_k)^{(a'_{i_k})}(x_{t_1})^{(a'_{t_1})} \cdots (x_{t_{m-k}})^{(a'_{t_{m-k}})} \equiv g \pmod{T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)}.$$

Thus, g is indeed a consequence of f modulo $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$. $\square$

Definition 5.5. We define a quasi-order in $\mathcal{B}_a$ by $f \preceq_{\mathcal{B}_a} g$ if $d(f) \leq d(g)$, for $f, g \in \mathcal{B}_a$.

Proposition 5.6. *The set $(\mathcal{B}_a, \preceq_{\mathcal{B}_a})$ has the finite basis property*

Proof. Suppose there is an infinite sequence $\{f_j\}_{j \geq 0}$ of pairwise incomparable elements in $\mathcal{B}_a$ with respect to the order $\preceq_{\mathcal{B}_a}$. According to the previous definition the last sequence induces the sequence $\{d(f_j)\}_{j \geq 0}$ in $D(\mathbb{N})$. The elements of the sequence $\{d(f_j)\}_{j \geq 0}$ are pairwise incomparable, but this contradicts the fact that $D(\mathbb{N})$ has the finite basis property. $\square$

Theorem 5.7. Any $T_{\mathbb{Z}}$ -ideal I containing $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ has a finitely generated as a $T_{\mathbb{Z}}$ -ideal, modulo the $T_{\mathbb{Z}}$ -ideal generated by $\{x^i \mid i \notin \{1, 2, \dots, n\}\}$.

Proof. If K is a finite field, then the identities $\{x^i \mid i \notin \{1, 2, \dots, n\}\}$ imply that the number of elements modulo $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n)$ is finite, and the result follows trivially. Therefore, let us assume that the ground field is infinite.

Since K is an infinite field, we know that I is generated as a $T_{\mathbb{Z}}$ -ideal by its multi-homogeneous polynomials. Hence, for each $a \in \{1, 2, \dots, n\}$, there exist subsets $\mathcal{B}'_a \subseteq \mathcal{B}_a$ such that

$$I = \langle \mathcal{B}'_a \mid a \in \{1, 2, \dots, n\} \rangle_{T_{\mathbb{Z}}} \mod T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n).$$

By Proposition 5.6 and Theorem 5.2, for each $a \in \{1, 2, \dots, n\}$, there exists $\mathcal{S}'_a \subseteq \mathcal{B}'_a$ such that $\mathcal{S}'_a$ is finite and $\mathcal{S}'_a \subseteq \mathcal{B}'_a \subseteq \bar{\mathcal{S}}'_a$. Consequently, given $g \in \mathcal{B}'_a$, there exists $f \in \mathcal{S}'_a$ such that $d(f) \leq d(g)$. By Lemma 5.4, $g \in \langle f \cup T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n) \rangle_{T_{\mathbb{Z}}}$, which implies

$$I = \langle \mathcal{S}'_a \mid a \in \{1, 2, \dots, n\} \rangle_{T_{\mathbb{Z}}} \mod T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^n).$$

Hence, the $T_{\mathbb{Z}}$ -ideal I , modulo the $T_{\mathbb{Z}}$ -ideal generated by $\{x^i \mid i \notin \{1, 2, \dots, n\}\}$, is finitely generated. $\square$

Corollary 5.8. For every i such that $n < i$, the T_i -ideal $T_i(\Gamma_n^i)$ has the Specht property.

5.2. Finite cyclic group

Let us first consider the situation of the nontrivial cyclic group. Again, such a group can be represented as $\mathbb{Z}_i$ for some positive integer i such that $i \leq n$. As in the previous subsection, we define the following sets:

$$\mathcal{B}_{\bar{a}} = \left\{ x_0^{\bar{a}}(x_1)^{(\alpha_1)} \dots (x_k)^{(\alpha_k)} \mid 1 < a \leq n, 0 \leq \sum_{j=1}^l \alpha_j \leq n - a, \alpha_i \geq 0, \|x_i\| = \bar{1}, \right. \\ \left. \text{for every } i \right\}.$$

Additionally, given a natural number k , for each $j = 1, \dots, k$, we define:

$$\mathcal{C}_j = \left\{ x_j(x_1)^{(\alpha_1)} \dots (x_k)^{(\alpha_k)} \mid 0 < \sum_{l=1}^k \alpha_l \leq n, \alpha_i \geq 0, \text{ for every } i \right\}. \quad (26)$$

Now, repeating once again what was done in the previous subsection, we can establish the following results.

Lemma 5.9. *Let f, g be elements in $\mathcal{B}_{\bar{a}}$ or in $\mathcal{C}_j$. If $d(f) \leq d(g)$, then g is a consequence of f modulo $T_i(\Gamma_i^n)$, that is $g \in \langle f \cup T_i(\Gamma_i^n) \rangle_{T_{\mathbb{Z}_i}}$.*

Definition 5.10. We define a quasi-order in $\mathcal{B}_{\bar{a}}$ by $f \preceq_{\mathcal{B}_{\bar{a}}} g$ if $d(f) \leq d(g)$, for $f, g \in \mathcal{B}_{\bar{a}}$. Similarly, we define a quasi-order on $\mathcal{C}_j$.

Proposition 5.11. *The set $(\mathcal{B}_{\bar{a}}, \preceq_{\mathcal{B}_{\bar{a}}})$ (as well as $(\mathcal{C}_j, \preceq_{\mathcal{C}_j})$) has the finite basis property.*

We now have all ingredients to establish the final result of this subsection.

Theorem 5.12. *For every i such that $i \leq n$, the T_i -ideal $T_i(\Gamma_i^n)$ has the Specht property.*

Proof. As it was mentioned in Theorem 5.7, we assume the ground field is infinite. This assumption is possible because the nilpotent identities given in Lemma 4.10 imply a finite number of elements in $\mathcal{L}_0^n$ modulo $T_{\mathbb{Z}}(\mathcal{L}_0^n)$, and the result follows trivially.

Let I be a T_i -ideal of $\mathcal{L}\langle X^{\mathbb{Z}_i} \rangle$ such that $T_i(\Gamma_i^n) \subseteq I$. Since K is an infinite field, I is generated as a $T_{\mathbb{Z}}$ -ideal by its multihomogeneous polynomials. Hence, for each $\bar{a} \in \{\bar{0}, \bar{1}, \dots, \overline{i-1}\}$, there exist subsets $\mathcal{B}'_{\bar{a}} \subseteq \mathcal{B}_{\bar{a}}$ such that

$$I = \langle \mathcal{B}'_{\bar{a}}, \prod_{j=1}^n \mathcal{C}'_j \mid a \in \{1, 2, \dots, n\} \rangle_{T_{\mathbb{Z}_i}} \mod T_i(\Gamma_i^n).$$

Here, $\mathcal{C}'_j$ denotes a subset of $\mathcal{C}_j$ such that $\prod_{j=1}^n \mathcal{C}'_j$ encompasses all graded identities in I with only variables of $\mathbb{Z}_i$ -degree $\bar{1}$.

By Proposition 5.11, Theorem 5.2 and 5.3, for each $a \in \{1, 2, \dots, n\}$, there exists $\mathcal{S}'_a \subseteq \mathcal{B}'_a$ ($\mathcal{F}' \subseteq \prod_{j=1}^n \mathcal{C}'_j$, respectively). These subsets are finite, and $\mathcal{S}'_a$ ($\mathcal{F}'$, respectively) satisfies $\mathcal{S}'_a \subseteq \mathcal{B}'_a \subseteq \mathcal{S}'_a$ ($\mathcal{F}' \subseteq \prod_{j=1}^n \mathcal{C}'_j \subseteq \mathcal{F}'$, respectively).

Given $g \in \mathcal{B}'_a$, there exists $f \in \mathcal{S}'_a$ such that $d(f) \leq d(g)$. Similarly, for each $g = \sum_{j=1}^n g_j \in \prod_{j=1}^n \mathcal{C}'_j$, there exists $f = \sum_{j=1}^n f_j \in \mathcal{F}'$ such that $d(f_j) \leq d(g_j)$ for each $j \in \{1, \dots, n\}$. By Lemma 5.9, $g \in \langle f \cup T_i(\Gamma_i^n) \rangle_{T_{\mathbb{Z}_i}}$, then

$$I = \langle \mathcal{S}'_a \cup \mathcal{F}' \mid a \in \{1, 2, \dots, n\} \rangle_{T_{\mathbb{Z}_i}} \mod T_i(\Gamma_i^n).$$

Thus, the $T_{\mathbb{Z}_i}$ -ideal I is finitely generated, and the result follows. $\square$

Now, let us consider the case of the trivial group. The proof of the Specht property remains largely unchanged from the previous case, with the only difference being the analysis over the set $\mathcal{C}_j$ as defined in (26), and only in it. By referencing the earlier structure of $\mathcal{C}_j$, the same techniques apply, demonstrating that the set satisfies the conditions required for the Specht property. The key idea is that the constraints on the degrees and indices in the set remain applicable even in the context of a trivial group. Thus, we can conclude that the Specht property holds in this setting as well. Since the

argument follows similar reasoning, we will simply state the result here, omitting the full proof for brevity.

Theorem 5.13. *The T -ideal $T(\mathcal{L}_0^n)$ has the Specht property.*

6. An analog of null-filiform Leibniz algebras in the infinite-dimensional case

We conclude this paper by studying an analog of null-filiform Leibniz algebras with infinite dimension. A pioneeristic discussion on this topic can be found in [31], while studies on ordinary identities over infinite fields are presented in [26]. In this context, we will focus on gradings and graded identities, expanding upon these foundational works.

In [31, Lemma 1], Omirov introduced and studied infinite-dimensional analogues of 0-filiform and filiform Leibniz algebras. More precisely, he showed that a potentially nilpotent infinite-dimensional Leibniz algebra, denoted here by $\mathcal{L}_0^\infty$, satisfying

$$\dim_K(\mathcal{L}_0^\infty)^{\langle i \rangle} / (\mathcal{L}_0^\infty)^{\langle i+1 \rangle} = 1,$$

for all i , admits a basis $\{e_1, e_2, e_3, \dots\}$ such that

$$[e_i, e_1] = e_{i+1}, \quad i \geq 1,$$

with all other products vanishing. This construction provides a concrete realization of an infinite-dimensional potentially nilpotent Leibniz algebra and closely follows the approach presented in [3, Lemma 1]. Consequently, if $\mathcal{L}_0^\infty$ is graded by a group, the element e_1 is always homogeneous. As a result, by following verbatim the proof of Theorem 3.3, we obtain the following result.

Lemma 6.1. *Let $\mathcal{L}_0^\infty$ be a potentially nilpotent Leibniz algebra (i.e. $\cap_{i=1}^\infty (\mathcal{L}_0^\infty)^{\langle i \rangle} = 0$) and let $\dim_K(\mathcal{L}_0^\infty)^{\langle i \rangle} / (\mathcal{L}_0^\infty)^{\langle i+1 \rangle} = 1$. Then, up to equivalence, the group gradings on $\mathcal{L}_0^\infty$ are the following:*

- (1) *The trivial grading given by $\mathcal{L}_0^\infty = \langle e_1, e_2, \dots \rangle$;*
- (2) *The $\mathbb{Z}$ -grading given by $\Gamma_{\mathbb{Z}}^\infty: \mathcal{L}_0^\infty = \langle e_1 \rangle_1 \oplus \langle e_2 \rangle_2 \oplus \dots$;*
- (3) *For any $i \leq 2$, the $\mathbb{Z}_i$ -grading given by $\Gamma_i^\infty: \mathcal{L}_0^\infty = L^{\bar{0}} \oplus L^{\bar{1}} \oplus \dots \oplus L^{\overline{i-1}}$, where each homogeneous subspace $L^{\bar{s}}$ is generated by elements e_j , such that $j \equiv s \pmod i$.*

Due to the explanations provided in the previous subsection, and closely mirroring the approach taken in Theorem 3.4, we can restate the main results from Section 4 as follows.

Theorem 6.2. *Over an infinite field and under the hypotheses of Lemma 6.1, we have:*

(1) The polynomials

$$x^i, \quad x_1^j x_2^k \quad \text{and} \quad x_1^1 x_2^1 - x_2^1 x_1^1,$$

where $i < 1$, $j \geq 1$ and $k \geq 2$, form a minimal basis for $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^{\infty})$, as a $T_{\mathbb{Z}}$ -ideal;

(2) The polynomials

$$x_1^{\bar{a}} x_2^{\bar{b}},$$

where $1 \leq a \leq i$, $2 \leq b \leq i$, form a minimal basis for the graded identities of $T_i(\Gamma_i^{\infty})$, as a T_i -ideal;

(3) The polynomial

$$x_1(x_2 x_3)$$

forms a minimal basis for $T(\mathcal{L}_0^{\infty})$, as a T -ideal;

Proof. The proof of item (1) is very similar to that of Proposition 4.3; therefore, it is omitted. For item (2), an analog of the result found in Lemma 4.11 applies, as we can restrict our consideration to multihomogeneous polynomials in $T_i(\Gamma_i^{\infty})$. Moreover, since $\mathcal{L}_0^{\infty}$ is not nilpotent, the identity (15) does not hold for this algebra. However, by carefully following the steps of the proof of Proposition 4.12, we can establish this item. The last item can be found in [26, Theorem 18]. $\square$

Theorem 6.3. Over a finite field K such that $|K| = q$ and under the hypotheses of Lemma 6.1, we have the following.

(1) The polynomials

$$x^i, \quad x_1^j x_2^k \quad \text{and} \quad x_1^1 x_2^1 - x_2^1 x_1^1,$$

where $i < 1$, $j \geq 1$ and $k \geq 2$, together with the identities $(x_0^1)^{(q)} x_1^1 - x_0^1 (x_1^1)^{(q)}$ and $x_0^k (x_0^1)^{(q)} x_1^1 - x_0^k x_0^1 (x_1^1)^{(q)}$ form a basis for $T_{\mathbb{Z}}(\Gamma_{\mathbb{Z}}^{\infty})$, as a $T_{\mathbb{Z}}$ -ideal.

(2) The polynomials

$$x_1^{\bar{a}} x_2^{\bar{b}},$$

and

$$x_0^{\bar{a}} (x_0^{\bar{1}})^{(q)} (x_1^{\bar{1}}) - x_0^{\bar{a}} (x_0^{\bar{1}}) (x_1^{\bar{1}})^{(q)},$$

where $1 \leq a \leq i$ and $2 \leq b \leq i$, form a basis for the graded identities of $T_i(\Gamma_i^{\infty})$, as a T_i -ideal.

(3) *The polynomial*

$$x_1(x_2x_3) \quad \text{and} \quad x_0x_1^{(q)}x_2 - x_0x_1x_2^{(q)}$$

form a minimal basis for $T(\mathcal{L}_0^\infty)$, as a T -ideal.

Proof. Concerning item (1), see Proposition 4.4. The proofs of items (2) and (3) are similar to those of Proposition 4.13 and Theorem 4.18, respectively, removing the condition of bounded nilpotency. $\square$

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Claudemir Fideles reports financial support was provided by State of Sao Paulo Research Foundation. Claudemir Fideles reports financial support was provided by National Council for Scientific and Technological Development. Luis Fortunani reports financial support was provided by Coordination for the improvement of Higher Education Personnel. Luis Fortunani reports financial support was provided by State of Sao Paulo Research Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors would like to thank the anonymous referee for her/his useful comments and suggestions.

Data availability

No data was used for the research described in the article.

References

- [1] L.E. Abanina, S.P. Mishchenko, The variety of Leibniz algebras defined by the identity $x(y(z t)) \equiv 0$, *Serdica Math. J.* 29 (2003) 291–300.
- [2] E. Aljadeff, A. Kanel-Belov, Y. Karasik, Kemer's theorem for affine PI algebras over a field of characteristic zero, *J. Pure Appl. Algebra* 220 (8) (2016) 2771–2808.
- [3] S.A. Ayupov, B.A. Omirov, On some classes of nilpotent Leibniz algebras, *Sib. Mat. Zh.* 42 (1) (2001) 18–29.
- [4] Y. Bahturin, *Identical Relations in Lie Algebras*, VNU Science Press, B.V., Utrecht, 1987.
- [5] Yu.A. Bakhturin, A.Yu. Ol'sanskii, Identical relations in finite Lie rings, *Mat. Sb. (N.S.)* 96 (138) (1975) 543–559 (in Russian); English Transl.: *Math. USSR Sb.* 25 (1975) 507–523.
- [6] S.A. Ayupov, B.A. Omirov, I.S. Rakhimov, *Leibniz Algebras: Structure and Classification*, CRC Press, Taylor & Francis Group, 2020.
- [7] D.W. Barnes, On Levi's theorem for Leibniz algebras, *Bull. Aust. Math. Soc.* 86 (2) (2012) 184–185.

- [8] A.M. Bloh, On a generalization of the concept of Lie algebra, *Dokl. Akad. Nauk SSSR* 165 (3) (1965) 471–473.
- [9] A.J. Calderón, L.M. Camacho, I. Kaygorodov, B. Omirov, Abelian groups gradings on null-filiform and one-parametric filiform Leibniz algebras, *J. Geom. Phys.* 170 (2021) 104384.
- [10] L. Centrone, C. Zargeh, Varieties of null-filiform Leibniz algebras under the action of Hopf algebras, *Algebr. Represent. Theory* 26 (2) (2023) 631–648.
- [11] D.M. Correa, P. Koshlukov, Specht property of varieties of graded Lie algebras, *Monatshefte Math.* 202 (1) (2023) 65–92.
- [12] V. Drensky, Identities in Lie algebras, *Algebra Log.* 13 (1974) 265–290 (Russian). English Translation: *Algebra Log.* 13 (1974) 150–165.
- [13] V. Drensky, G.M.P. Cattaneo, Varieties of metabelian Leibniz algebras, *J. Algebra Appl.* 1 (1) (2002) 31–50.
- [14] A. Elduque, M. Kochetov, Gradings on simple Lie algebras, *Mathematical Surveys and Monographs*, vol. 189, AMS, Providence, RI, 2013. Atlantic Association for Research in the Mathematical Sciences (AARMS), Halifax, NS.
- [15] A. Giambruno, M. Zaicev, Polynomial Identities and Asymptotic Methods, *Math. Surveys Monogr.*, vol. 122, Amer. Math. Soc., Providence, RI, 2005.
- [16] A. Giambruno, M. Zaicev, S. Mishchenko, Group actions and asymptotic behavior of graded polynomial identities, *J. Lond. Math. Soc.* (2) 66 (2) (2002) 259–312.
- [17] A.V. Grishin, Examples of T-spaces and T-ideals of characteristic 2 without the finite basis property, *Fundam. Prikl. Mat.* 5 (1) (1999) 101–118 (Russian).
- [18] G. Higman, Ordering by divisibility in abstract algebras, *Proc. Lond. Math. Soc.* 3 (2) (1952) 326–336.
- [19] A. Iltiyakov, On finite basis of identities of Lie algebra representations, *Nova J. Algebra Geom.* 1 (3) (1992) 207–259.
- [20] A.Ya. Kanel-Belov, Counterexamples to the Specht problem, *Mat. Sb.* 191 (2000) 13–24 (Russian). English Translation: *Sb. Math.* 191 (2000) 329–340.
- [21] A.R. Kemer, Ideals of identities of associative algebras, *AMS Translations of Math. Monographs*, vol. 87, AMS, Providence, RI, 1991.
- [22] R.L. Kruse, Identities satisfied by a finite ring, *J. Algebra* 26 (1973) 308–318.
- [23] J.-L. Loday, Une version non commutative des algèbres de Lie: les algèbres de Leibniz, *Enseign. Math.* 39 (1993) 269–293 (French) [A noncommutative version of Lie algebras: the Leibniz algebras].
- [24] J.-L. Loday, T. Pirashvili, Universal enveloping algebras of Leibniz algebras and (co)homology, *Math. Ann.* 296 (1993) 139–158.
- [25] I.V. Lvov, Varieties of associative rings, *Algebra Log.* 12 (3) (1973) 269–297 (Russian). English transl.: *Algebra Log.* 12 (3) (1973) 150–167.
- [26] T.C. de Mello, M.S. Souza, Polynomial identities and images of polynomials on null-filiform Leibniz algebras, *Linear Algebra Appl.* 679 (2023) 246–260.
- [27] S. Mishchenko, A. Valenti, A Leibniz variety with almost polynomial growth, *J. Pure Appl. Algebra* 202 (2005) 82–101.
- [28] S. Mishchenko, A. Valenti, The Leibniz algebras of almost polynomial growth with the identity $x(y(zx)) \equiv 0$, *Vestnik Moskov. Univ. Ser. 1. Mat. Mekh.* 3 (2010) 18–23.
- [29] A. Naurazbekova, U. Umirbaev, Identities of dual Leibniz algebras, *TWMS J. Pure Appl. Math.* 1 (1) (2010) 86–91.
- [30] S. Oates, M. Powell, Identical relations in finite groups, *J. Algebra* 1 (1964) 11–39.
- [31] B.A. Omirov, Thin Leibniz algebras, *Math. Notes* 80 (2006) 244–253.
- [32] V.V. Shchigolev, Construction of non-finitely based T-ideals, *Commun. Algebra* 29 (9) (2001) 3935–3941 (Special issue dedicated to Alexei Ivanovich Kostrikin).
- [33] M.R. Vaughan-Lee, Abelian by nilpotent varieties, *Q. J. Math. Oxf. Ser.* (2) 21 (1970) 193–202.
- [34] K.A. Zhevlakov, A.M. Slin’ko, I.P. Shestakov, A.I. Shirshov, *Rings That Are Nearly Associative*, Academic Press, 1982, ISBN: 0-12-779850.